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Grothendieck Topologies on the Category of Finite Probability Spaces

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1 Abstract

The category of finite probability spaces lacks the formation of categorical products and is therefore not Cartesianly closed. It can be enlarged to a category of sheaves using the notion of Grothendieck topologies. In this thesis, we give a large class of examples of non-trivial Grothendieck topologies on the category of finite probability spaces, namely the projection topology, the X-projection topology and combinations of them. We then prove that each of these are subcanonical by first showing a far more powerful result that the atomic topology on the category of finite probability spaces is subcanonical. Additionally, we give a rigorous treatise regarding the relevant structure of the category of finite probability spaces and give a characterization of isomorphisms in the category.

Contents

2 Introduction

The category of finite probability spaces **Prob** has been proven to be a useful tool in analyzing mathematical fields such as information theory and tropical probability theory. For example, in the papers [1] and [2], J. W. Portegies & Matveev R utilized **Prob** in deriving information inequalities for information theory.

However, the category **Prob** does lack certain structure. For example, **Prob** lacks the formation of categorical products. With this in mind, **Prob** is not a Cartesian closed category, which is a property providing a framework for computations in a convenient way.

The need for a Cartesian closed category for probability theory has been pointed out in the article of C. Heunen et. al. [3]. The article deals with the category of measurable spaces **Meas** not being Cartesian closed, by formalizing probability theory in the category of so called quasi-Borel spaces **QBS**. This category is Cartesian closed and there is an embedding

$$\mathbf{Meas} \hookrightarrow \mathbf{QBS}$$

where **Meas** is the category of measurable spaces. The theory of finite probability spaces can then be conducted inside **QBS**.

A similar situation for manifold theory has been worked out in the book *Models for smooth infinitesimal analysis* by I. Moerdijk & G. E Reyes [4]. The category of smooth manifolds **Man** is, like **Prob**, not a Cartesian closed category. The book works around this problem by embedding **Man** into its presheaf category $\mathrm{Psh}(\mathbf{Man}) := [\mathbf{Man}^{\mathrm{op}}, \mathbf{Set}]$, which consists of the contravariant functors $\mathbf{Man}^{\mathrm{op}} \to \mathbf{Set}$. The embedding is given by the Yoneda embedding:

$$\begin{aligned} \mathbf{y} :\ \mathbf{Man} &\hookrightarrow \mathrm{Psh}(\mathbf{Man}), \\ M &\mapsto \mathrm{Hom}_{\mathbf{Man}}(-, M) \end{aligned}$$

and an object of Psh(**Man**) is viewed as a generalized manifold.

However, it is early on noted that this embedding **y** does not preserve all the properties that **Man** possesses. For example, open covers are not preserved by **y** and as a result, the generalised unit interval $\mathbf{y}([0,1]) \in \mathrm{Psh}(\mathbf{Man})$ is not compact when treated as a generalized manifold, but $[0,1] \in \mathbf{Man}$ is compact. The problem lies in the fact that there are too many objects in Psh(**Man**).

The book works around this problem by restricting the presheaf category cleverly. It endows **Man** with a Grothendieck topology J and the theory of manifolds is then conducted in the J-sheaf category $\mathrm{Sh}_J(\mathbf{Man})$. The category $\mathrm{Sh}_J(\mathbf{Man})$ is a full subcategory of Psh(**Man**) and is always cartesian closed. The Grothendieck topology J controls what presheaves we want to keep in $\mathrm{Sh}_J(\mathbf{Man})$ and which ones we throw away: larger Grothendiek topologies give less sheaves.

So instead of conducting the theory of finite probability spaces in **QBS**, like the aforementioned article [3], inspired by the book models for smooth infinitesimal analysis ([4]) we could try to enlarge **Prob** to a J-sheaf category $\mathrm{Sh}_J(\mathbf{Prob})$ for some Grothendieck topology J, then view an object of $\mathrm{Sh}_J(\mathbf{Prob})$ as a generalized finite probability space and perhaps derive useful results or tools in information theory or tropical probability theory. It would also be interesting to see if the theory of finite probability spaces in $\mathrm{Sh}_J(\mathbf{Prob})$ would help in formalizing probability theory for digital proof assistants like CoQ.

So which Grothendieck topology J do we endow on **Prob**? From the book *Sheaves in Geometry and Logic* by S. MacLane & I. Moerdijk [5], we already get some examples of Grothendieck topologies we can endow on any category: the trivial topology J_{triv}, the discrete topology J_{d} and the dense topology J_{de}. The goal of this project was to find a Grothendieck topology J on **Prob**, different from the three we have previously mentioned, such that the category of J-sheaves $\mathrm{Sh}_J(\mathbf{Prob})$ still enlarges **Prob** i.e. we still have an embedding

$$\mathbf{Prob} \hookrightarrow \mathrm{Sh}_J(\mathbf{Prob}).$$

If we have such an embedding, the Grothendieck topology J is called subcanonical and the largest of such subcanonical Grothendieck topology is called the canonical topology.

By utilizing the theory of coverages as presented in the book *Sketches of an Elephant* by P. T. Johnstone [6], we found a large class of non-trivial, subcanonical Grothendieck topologies on **Prob** called the X-projection topologies and the projection topologies. Whilst proving these Grothendieck topologies were subcanonical on **Prob**, we realised that we could prove a far more powerful result: the aforementioned dense topology J_{de} is a subcanonical Grothendieck topology on **Prob**. Moreover, it turns out that J_{de} is actually the canonical topology on **Prob**.

With these results, we lay a foundation for the generalised probability theory that can be conducted in the category of J_{de}-sheaves $\mathrm{Sh}_{J_{\mathrm{de}}}(\mathbf{Prob})$, and possibly derive useful results in information theory, tropical probability theory and formalising probability theory for proof assistants.

2.1 Outline of the report

We first establish some properties of **Prob**, which are stated in the previously mentioned [1], which will be useful later: isomorphisms are classified, a proof that **Prob** does not admit all products is given and we give some examples of finite probability spaces. Section 4 gives the necessary notions of the book *Sheaves in Geometry and Logic* by S. MacLane & I. Moerdijk [5] and meant for us to understand the general theory of sheaves and Grothendieck topologies. Section 5 answers the main question in that we define a class of Grothendieck topologies on **Prob**, namely the X-projection topology J_X for any finite probability space X and the projection topology J_{proj}. We will also explore combinations of these Grothendieck topologies. In section 6, we turn to a popular example of a Grothendieck topology which is the atomic topology J_{at}. It turns out that J_{at} enjoys useful properties that are later utilized to prove not only that we have embeddings $\mathbf{Prob} \hookrightarrow \mathrm{Sh}_{J_X}(\mathbf{Prob})$ and $\mathbf{Prob} \hookrightarrow \mathrm{Sh}_{J_{\mathrm{proj}}}(\mathbf{Prob})$, but we classify which Grothendieck topologies J on **Prob** allow an embedding $\mathbf{Prob} \hookrightarrow \mathrm{Sh}_J(\mathbf{Prob})$.

3 The category Prob

In this section we will expand and rigorously define and describe properties of **Prob**, which is a category utilized in the the paper [1]. We will define the category **Prob** properly and investigate its properties to serve as a foundation for the rest of this report. The isomorphisms in this category can be characterized and we will give criteria which is can be used to conclude if two spaces are isomorphic to each other or not. We will use this later to prove that **Prob** does not admit products. To conclude the section, we want to investigate the size of **Prob**.

3.1 Reductions

As previously mentioned, the category in which we will work will be the category of finite probability spaces (X, p_X) where X is a set with finite support with respect to a probability measure $p_X : 2^X \to [0, 1]$ which we will call the *probability measure* of X. This in particular means that

- $p_X(\emptyset) = 0$ and $p_X(X) = 1$;
- for every countable collection of subsets $(A_i)_{i\in\mathbb{N}} \subseteq X$ of pairwise disjoint sets,

$$p_X(\bigcup_{i\in\mathbb{N}} A_i) = \sum_{i\in\mathbb{N}} p_X(A_i).$$

If we have an element $x \in X$, then we denote by $p_X(x) := p_X(\{x\})$.

It is worth noting that to define a probability space, we need a sigma algebra. We will always choose the discrete sigma algebra and will not mention it in the rest of the report.

Definition 3.1.1. *A finite probability space (X, p_X) is a pair with X a set, p_X a probability measure such that the set* $\mathrm{supp}(X, p_X) := \{x \in X \mid p_X(x) > 0\}$*, which we will call the support of X, is finite.*

Definition 3.1.2 ([1])**.** *Let (X, p_X) and (Y, p_Y) be finite probability spaces. A function $f : X \to Y$ is called measure-preserving if*

$$f_\# p_X = p_Y,$$

where $f_\#$ is the pushforward measure.

We first note that since we work with finite probability spaces, that for all $A \subseteq Y$

$$(f_\# p_X)(A) = p_X(f^{-1}(A)) = \sum_{x\in f^{-1}(A)} p_X(x),$$

by definition of the probability measure p_X.

Lemma 3.1.3. *measure-preserving maps are closed under composition.*

Proof. Let (X, p_X), (Y, P_Y) and (Z, p_Z) be finite probability spaces and $f : X \to Y$ and $g : Y \to Z$ measure-preserving. Then for all $A \subseteq Z$,

$$\begin{aligned}(g \circ f)_\#(p_X)(A) &= p_X((g \circ f)^{-1}(A)) = p_X(f^{-1}(g^{-1}(A))) \\ &= (f_\# p_X)(g^{-1}(A)) = f_\# g_\# p_X(A) \\ &= g_\# p_Y(A) = p_Z(A).\end{aligned}$$

So $g \circ f : X \to Z$ is a measure-preserving map. □

Lemma 3.1.4. *The identity map is measure-preserving.*

Proof. Let (X, p_X) be a finite probability space. Then for all $A \subseteq X$

$$(\mathrm{id}_X)_\# p_X(A) = p_X(\mathrm{id}_X^{-1}(A)) = p_X(A).$$

So the identity is a measure-preserving map. □

Lemma 3.1.5. *Let (X, p_X) and (Y, p_Y) be finite probability spaces. Let $f : X \to Y$ be a measure-preserving map such that f is also bijective. Then the inverse of f is also measure-preserving.*

Proof. Denote by f^{-1} the inverse of f. Then we see for all subsets $A \subseteq X$,

$$\begin{aligned} f^{-1}_\# p_Y(A) &= p_Y((f^{-1})^{-1}(A)) \\ &= p_Y\{y \in Y \mid f^{-1}(y) \in A\}) \\ &= p_Y\{y \in f(X) \mid f^{-1}(y) \in A\}) \\ &= p_Y(f(A)) = f_\# p_X(f(A)) \\ &= p_X(f^{-1}f(A)) = p_X(A), \end{aligned}$$

where we used in the last equality that f is injective since it is bijective. Hence the inverse is measure-preserving. □

We will now define what morphisms in the category of finite probability spaces will be.

Definition 3.1.6 ([1]). *Let (X, p_X) and (Y, p_Y) be two finite probability spaces and let $f, g : X \to Y$ be measure-preserving. We call f and g equivalent they agree on the support of (X, p_X), i.e.*

$$f|_{\mathrm{supp}(X,p_X)} = g|_{\mathrm{supp}(X,p_X)}. \tag{1}$$

We call the equivalence class of such functions a reduction and denote it by $[f] : (X, p_X) \to (Y, p_Y)$.

We start by noting that since a reduction $[f] : (X, p_X) \to (Y, p_Y)$ is an equivalence class of measure-preserving maps it makes no sense to evaluate on an element $x \in X$. However, since two representatives of $[f]$, say f and g, agree on elements of X with positive measure, we will by abuse of notation evaluate on $[f]$, i.e.

$$[f](x) := f(x)$$

for all $x \in X$ with $p_X(x) > 0$ and f any representative of $[f]$. Using this, we can check equality of two reductions easily.

To form a category using reductions and finite probability spaces, we need a form of composition. As we are working with equivalence classes of maps, this is not as trivial as working with standard functions.

Lemma 3.1.7. *Let (X, p_X), (Y, P_Y) and (Z, p_Z) be finite probability spaces and $[f] : (X, p_X) \to (Y, p_Y)$ and $[g] : (Y, p_Y) \to (Z, p_Z)$ be reductions. Then $[g \circ f] : (X, p_X) \to (Z, p_Z)$ is also a reduction. Moreover, this equivalence class is independent of the of the choice of representatives f and g and is called the composition of $[f]$ and $[g]$ and will be denoted by,*

$$[g] \circ [f] := [g \circ f].$$

Proof. Let f and g be two representatives for $[f]$ and $[g]$ respectively. Then since f and g are measure-preserving, by Lemma 3.1.3, $g \circ f$ is measure-preserving, so the equivalence class $[g \circ f]$ is a reduction. We now show that $[g \circ f]$ is independent of the choice of the representatives. So let f and f' be two representatives of $[f]$ and g and g' be two representatives for $[g]$. Then by definition

$$f|_{\mathrm{supp}(X,p_X)} = f'|_{\mathrm{supp}(X,p_X)} \text{ and } g|_{\mathrm{supp}(Y,p_Y)} = g'|_{\mathrm{supp}(Y,p_Y)}.$$

We need to show that $g \circ f$ is equivalent to $g' \circ f'$ which means that they need to agree on the set $\text{supp}(X, p_X)$. Since f is measure-preserving, we see that for all $x \in X$ such that $p_X(x) > 0$, i.e., for all $x \in \text{supp}(X, p_X)$, that

$$p_Y(f(x)) = f_\# p_X(f(x)) > 0,$$

since $x \in f^{-1}(f(x))$. So $f(x) \in \text{supp}(Y, p_Y)$. So then we see for all $x \in \text{supp}(X, p_X)$,

$$(g \circ f)(x) = (g \circ f')(x) = (g' \circ f')(x),$$

so $(g \circ f)|_{\text{supp}(X,p_X)} = (g' \circ f')|_{\text{supp}(X,p_X)}$. So $g \circ f$ is equivalent to $g' \circ f'$. $\square$

Lemma 3.1.8. *There is a unique identity reduction for every finite probability space (X, p_X), i.e. some reduction $\text{id}_{(X,p_X)} : (X, p_X) \to (X, p_X)$ such that for all other reductions $[f] : (X, p_X) \to (Y, p_Y)$ and $[g] : (Z, p_Z) \to (X, p_X)$*

$$[f] \circ \text{id}_{(X.p_X)} = [f] \textit{ and } \text{id}_{(X.p_X)} \circ [g] = [g].$$

Proof. By Lemma 3.1.7, we see by choosing $\text{id}_{(X,p_X)} = [\text{id}_X]$ that

$$[f] \circ \text{id}_{(X.p_X)} = [f] \circ [\text{id}_X] = [f \circ \text{id}_X] = [f],$$

and

$$\text{id}_{(X.p_X)} \circ [g] = [\text{id}_X] \circ [g] = [\text{id}_X \circ g] = [g].$$

We now need to show this is identity is unique, let $\text{id}'_{(X,p_X)}$ be another identity reduction on the finite probability space (X, p_X). Then

$$\text{id}'_{(X,p_X)} = \text{id}'_{(X,p_X)} \circ \text{id}_{(X,p_X)} = \text{id}_{(X,p_X)},$$

which implies the identity is unique. $\square$

3.2 Isomorphisms

Now that we have properly defined reductions and proven some elementary properties of reductions, we can define the category **Prob**.

Definition 3.2.1 ([1]). *We denote by* **Prob** *the category with:*

- $\text{Ob}(\textbf{\textit{Prob}}) := \{(X, p_X) \mid X \textit{ is a set with } \text{supp}(X, p_X) \textit{ finite, and } p_X \textit{ is a probability measure on } X\}.$
- *For each* $(X, p_X), (Y, p_Y) \in \text{Ob}(\textbf{Prob})$,
  $\text{Hom}_{\textbf{Prob}}((X, p_X), (Y, p_Y)) := \{f : (X, p_X) \to (Y, p_Y) \mid f \textit{ is a reduction}\}.$
- *Composition is defined as the composition in Lemma 3.1.7,*
- *For all finite probability spaces (X, p_X), the identity is defined as in Lemma 3.1.8.*

If confusion is not likely, for all finite probability spaces (X, p_X) in **Prob** *by abuse of notation, we will omit the probabity distribution in the notation, so we will denote X for the finite probability space (X, p_X).*

We now aim to find a characterization of isomorphisms in **Prob**.

Lemma 3.2.2. *Let X and Y be sets and suppose $f : X \to Y$ is an injective map between them. Then for every $x \in X$, we have $\{x\} = f^{-1}(f(x))$.*

Proof. Let $x \in X$, then

$$\begin{aligned} f^{-1}(f(x)) &= \{y \in X \mid f(y) \in \{f(x)\}\} \\ &= \{y \in X \mid f(y) = f(x)\} \\ &= \{y \in X \mid y = x\} \\ &= \{x\}, \end{aligned}$$

which completes the proof. □

Lemma 3.2.3. *Let X and Y be finite probability spaces. Suppose we have a measure-preserving injective map $f : X \to Y$. Then for every $x \in X$, we have $p_X(x) = p_Y(f(x))$.*

Proof. Let $x \in X$. Then since f is injective and by Lemma 3.2.2,

$$p_X(x) = p_X(f^{-1}f(x)) = f_{\#}p_X(f(x)) = p_Y(f(x)),$$

which completes the proof. □

Definition 3.2.4. *Let X be a finite probability space. We define the support space of X by* $\text{supp}(X) := (\text{supp}(X), p_{\text{supp}(X)})$ *with* $p_{\text{supp}(X)}(A) = p_X(A)$ *for every subset* $A \subseteq \text{supp}(X)$.

Lemma 3.2.5. *For every finite probability space X, the support space* $\text{supp}(X)$ *is a finite probability space.*

Proof. We simply have

$$p_{\text{supp}(X)}(\text{supp}(X)) = p_X(\text{supp}(X)) = 1$$

which shows $p_{\text{supp}(X)}$ is a probability measure. □

Lemma 3.2.6. *Any finite probability space is isomorphic to its support space.*

Proof. Let X be a finite probability space. We need to find two reductions $f : X \to \text{supp}(X)$ and $g : \text{supp}(X) \to X$ such that $f \circ g = \text{id}_{\text{supp}(X)}$ and such that $g \circ f = \text{id}_X$. Define for every x with positive measure, $f(x) = x$ and $g(x) = x$. Then we have

$$\begin{aligned} f \circ g &= \text{id}_{\text{supp}(X)} \\ g \circ f &= \text{id}_X. \end{aligned}$$

So $X \simeq \text{supp}(X)$. □

Proposition 3.2.7. *Let X and Y be two finite probability spaces. Then $X \simeq Y$ if and only if there is a bijective measure-preserving map between* $\text{supp}(X)$ *and* $\text{supp}(Y)$.

Proof. Suppose we have a bijective measure-preserving map $f : \text{supp}(X) \to \text{supp}(Y)$. Then since f is bijective, we have an inverse of f which we will denote by f^{-1}. By Lemma 3.1.5, f^{-1} is also measure-preserving. Then we have

$$\begin{aligned} [f] \circ [f^{-1}] &:= [f \circ f^{-1}] = \text{id}_{\text{supp}(Y)} \\ [f^{-1}] \circ [f] &:= [f^{-1} \circ f] = \text{id}_{\text{supp}(X)}. \end{aligned}$$

Then by Lemma 3.2.6, $X \simeq \text{supp}(X) \simeq \text{supp}(Y) \simeq Y$.
Suppose $X \simeq Y$, then by Lemma 3.2.6, we have that $\text{supp}(X) \simeq X \simeq Y \simeq \text{supp}(Y)$. So we have two reductions $f : \text{supp}(X) \to \text{supp}(Y)$ and $g : \text{supp}(Y) \to \text{supp}(X)$ such that they are inverses of each other. Then as every element in $\text{supp}(X)$ has positive measure, there is a unique representative of f and a unique representative of g, which are both measure-preserving and inverses of each other. So we have a bijective measure-preserving map between $\text{supp}(X)$ and $\text{supp}(Y)$. □

Corollary 3.2.8. *Let X and Y be finite probability spaces and let $f : \operatorname{supp}(X) \to Y$ be an injective measure-preserving map. Then $X \simeq Y$.*

Proof. By Lemma 3.2.6, it suffices to show that $\operatorname{supp}(X) \simeq \operatorname{supp}(Y)$. Now since $f : \operatorname{supp}(X) \to Y$ is an injective measure-preserving map, by Lemma 3.2.3 we have that for every $x \in \operatorname{supp}(X)$, that $p_X(x) > 0$ so also that $p_Y(f(x)) = p_Y(f(x)) > 0$. Therefore

$$\operatorname{im}(f) \subseteq \operatorname{supp}(Y).$$

By Proposition 3.2.7, it suffices to show that that f is bijective, and since it is already injective, it suffices to show that it is surjective. Let $y \in \operatorname{supp}(Y)$. Then

$$0 < p_Y(y) = (f)_\# p_X(y) = p_X(f^{-1}(y))$$

which implies $f^{-1}(y)$ is nonempty. So we can choose an $x \in \operatorname{supp}(X)$ such that $f(x) = y$ and therefore we see that f is surjective and hence bijective. So we have a bijective measure-preserving map between $\operatorname{supp}(X)$ and $\operatorname{supp}(Y)$ and therefore, by Proposition 3.2.7 we have that $X \simeq Y$. □

3.3 Products

We would now like to investigate if the category **Prob** admits categorical products. To do this we will exploit some structure of **Prob** namely, we can form the (non-categorical) product of two finite probability spaces. We emphasise that this is not a product in the categorical sense as it turns out that this object does not satisfy the universal property of the categorical product.

Definition 3.3.1 ([1]). *Let X and Y be two objects of **Prob**. We define $X \otimes Y$ to be the product of X and Y where the underlying set is the cartesian product $X \times Y$ and the product probability measure is given by*

$$(p_X \otimes p_Y)(x, y) := p_{X\times Y}(x, y) := p_X(x)p_Y(y)$$

*for every $(x, y) \in X \times Y$. We also define the n fold product of an object X of **Prob** to be*

$$X^n := \bigotimes_{i=1}^{n} X.$$

Notice that for all $A \subseteq X \times Y$,

$$(p_X \otimes p_Y)(A) = \sum_{(x,y)\in A} p_X(x)p_Y(y).$$

Definition 3.3.2. *Let X and Y be two finite probability spaces. Then we define the map $\pi_X : X \otimes Y \to X$, by*

$$\pi_X(x, y) := x$$

for every $(x, y) \in X \otimes Y$. We call this map the projection.

Lemma 3.3.3. *Projections are measure-preserving.*

Proof. Notice that for every $A \subseteq X$ we have

$$\begin{aligned}\pi_X^{-1}(A) &= \{(x, y) \in X \times Y \mid \pi_X(x, y) = x \in A\} \\ &= A \times Y.\end{aligned}$$

So then

$$\begin{aligned}(\pi_X)_\# p_{X\times Y}(A) &= p_{X\times Y}(\pi_X^{-1}(A)) = p_{X\times Y}(A \times Y) = p_X(A)p_Y(Y) \\ &= p_X(A),\end{aligned}$$

since $p_Y(Y) = 1$, for p_Y is a probabiltiy measure. The proof for π_Y is exactly the same. □

For **Prob** to admit all products, we need that for all X, Y finite probability spaces, we have another finite probability space Q with two reductions $\pi_X : Q \to X$ and $\pi_Y : Q \to Y$ such that for all other finite probability space O with reductions $f : O \to X$ and $g : O \to Y$, there is a unique reduction $\psi : O \to Q$ such that the diagram

$$\begin{tikzcd} & O \arrow[dl, bend right, "g"'] \arrow[d, dashed, "\psi"] \arrow[dr, bend left, "f"] & \\ X, & Q \arrow[l, "\pi_X"] \arrow[r, "\pi_Y"'] & Y \end{tikzcd}$$

commutes.

Theorem 3.3.4. *The category **Prob** does not admit all products.*

Proof. We will first prove that if a product Q for two finite probability spaces X_1 and X_2 exists, then $Q \simeq X_1 \otimes X_2$. We will then arrive at a contradiction by using properties of the probability measure of Q.
Suppose **Prob** does admit all products. Consider finite probability spaces X_1 and X_2. Then since we assumed **Prob** admits all products, we have another finite probability space Q with two reductions $\pi_{X_1} : Q \to X_1$ and $\pi_{X_2} : Q \to X_2$ such that for all other finite probability spaces O with two reductions $f : O \to X_1$ and $g : O \to X_2$, we have a unique reduction $\psi : O \to Q$ such that the diagram

$$\begin{tikzcd} & O \arrow[dl, bend right, "f"'] \arrow[d, dashed, "\psi"] \arrow[dr, bend left, "g"] & \\ X_1 & Q \arrow[l, "\pi_{X_1}"] \arrow[r, "\pi_{X_2}"'] & X_2 \end{tikzcd}$$

commutes. We choose the following diagram

$$\begin{tikzcd} & X_1 \otimes X_2 \arrow[dl, bend right, "q_1"'] \arrow[d, dashed, "\psi"] \arrow[dr, bend left, "q_2"] & \\ X_1 & Q \arrow[l, "\pi_{X_1}"] \arrow[r, "\pi_{X_2}"'] & X_2 \end{tikzcd}$$

where $X_1 \otimes X_2$ is the finite probablity space defined in 3.3.1 and q_1, q_2 are the equivalence classes of the usual set-theoretical projections, i.e.

$$q_i(x) = x_i, \tag{2}$$

for $i = 1, 2$ and all $x \in X_1 \otimes X_2$ with positve measure. So this is a product diagram in **Prob**.
We now aim to show there is an injective map $\psi' : \text{supp}(X_1 \otimes X_2) \to \text{supp}(Q)$. Let ψ' be a representative of the reduction ψ. Suppose $x, y \in X_1 \otimes X_2$ with $q_{X_1 \otimes X_2}(x) > 0$ and $q_{X_1 \otimes X_2}(y) > 0$ such that $\psi'(x) = \psi'(y)$. Then by commutativity of the diagram, for $i = 1, 2$,

$$x_i = q_i(x) = \pi_{X_i}(\psi'(x)) = \pi_{X_i}(\psi'(y)) = q_i(y) = y_i$$

so $x = y$ which implies ψ' is an injective measure-preserving map on the support of $X_1 \otimes X_2$ and Q. Then by Corollary 3.2.8, we have $Q \simeq X_1 \otimes X_2$. Now since Q with the reductions π_{X_1}, π_{X_2} is the product of (X_1, q_{X_1}) and (X_2, q_{X_2}) and since products are preserved under isomorphisms (see [7]), we see that $X_1 \otimes X_2$ with the reductions $\pi_{X_1} \circ \psi$, $\pi_{X_2} \circ \psi$, is also a categorical product of X_1 and X_2. Let Y be a finite probability space with reductions $f_i : Y \to X_i$. Then by the universal property of the product we get a unique reduction $\tau : Y \to X_1 \otimes X_2$ such that $(\pi_{X_i} \circ \psi) \circ \tau = f_i$. This is visualised in the commutative diagram

$$\begin{tikzcd}
& Y \arrow[ld, bend right, "f_1"'] \arrow[d, dashed, "\tau"] \arrow[rd, bend left, "f_2"] & \\
X_1 & X_1 \otimes X_2 \arrow[l, "\pi_{X_1}\circ\phi"] \arrow[r, "\pi_{X_2}\circ\phi"'] & X_2
\end{tikzcd}$$

Now for every subset U of $X_1 \times X_2$ of the form $U = A_1 \times A_2$ we have the following

$$\begin{aligned}
p_{X_1 \otimes X_2}(U) &= \tau_\# p_Y(U) = p_Y(\tau^{-1}(U)) \\
&= p_Y(\{y \in Y \mid \tau(y) \in U\}) \\
&= p_Y(\{y \in Y \mid \tau(y) \in A_1 \times A_2\}) \\
&= p_Y(\{y \in Y \mid \forall i \in \{1,2\},\ (\tau(y))_i \in A_i\}) \\
&= p_Y(\{y \in Y \mid \forall i \in \{1,2\},\ q_i(\tau(y)) \in A_i\}) \\
&= p_Y(\{y \in Y \mid \forall i \in \{1,2\},\ (\pi_{X_i} \circ \psi)(\tau(y)) \in A_i\}) \\
&= p_Y(\{y \in Y \mid \forall i \in \{1,2\},\ f_i(y) \in A_i\}) \\
&= p_Y(f_1^{-1}(A_1) \cap f_2^{-1}(A_2)).
\end{aligned}$$

This shows in particular that $p_Y(f_1^{-1}(A_1) \cap f_2^{-1}(A_2))$ is independent of the choice of the reductions f_1 and f_2.
We are now ready to arrive at a contradiction. Define $X = \{0,1\}$ with probablity measure $p_X(x) = \frac{1}{2}$. We also define $f : X \to X$ by

$$f : 0 \mapsto 1$$

$$f : 1 \mapsto 0.$$

Then f is a reduction. Indeed, for $x = 0, 1$

$$f_\# p_X(x) = p_X(f^{-1}(x)) = \frac{1}{2} = p_X(x).$$

Now define $U = \{(0,1)\} = \{0\} \times \{1\}$. Then this is a subset of $X \times X = \{0,1\} \times \{0,1\}$. Then if we choose $f_1, f_2 = \mathrm{id}_X$, we have that

$$\begin{aligned}
p_Q(U) &= p_X(f_1^{-1}(\{0\}) \cap f_2^{-1}(\{1\})) = p_X(\mathrm{id}_X^{-1}(\{0\}) \cap \mathrm{id}_X^{-1}(\{1\})) \\
&= p_Q(\{0\} \cap \{1\}) = p_Q(\emptyset) = 0.
\end{aligned}$$

If we choose however $f_1 = f$ and $f_2 = \mathrm{id}_X$, we get

$$\begin{aligned}
0 = p_Q(U) &= p_X(f^{-1}(\{0\}) \cap \mathrm{id}_X^{-1}(\{1\})) \\
&= p_X(\{1\} \cap \{1\}) = \frac{1}{2},
\end{aligned}$$

which is a contradiction. □

3.4 Examples of spaces and morphisms

This section is meant to introduce some finite probability spaces and morphisms that will become important in proofs in later chapters.

Lemma 3.4.1. *Let $\{X_i\}_{i=1}^n$ and $\{Y_i\}_{i=1}^n$ be two finite collections of finite probability spaces and for every $i \in \{1, ..., n\}$ let $f_i : X_i \to Y_i$ be a measure-preserving map. Then the map $(f_1, ..., f_n) : \bigotimes_{i=1}^n X_i \to \bigotimes_{i=1}^n Y_i$ defined by $(x_1, ..., x_n) \mapsto (f_1(x_1), ..., f(x_n))$ is measure-preserving.*

Proof. Let $(y_1, ..., y_n) \in \bigotimes_{i=1}^n Y_i$. Then notice that we have

$$\begin{aligned}(f_1, ..., f_n)^{-1}(y_1, ..., y_n) &= \{(x_1, ..., x_n) \in \prod_{i=1}^n X_i \mid (f_1, ..., f_n)(x_1, ..., x_n) = (y_1, ..., y_n)\} \\ &= \{(x_1, ..., x_n) \in \prod_{i=1}^n X_i \mid \forall i \in \{1, ..., n\},\ f_i(x_i) = y_i\} \\ &= \prod_{i=1}^n f_i^{-1}(y_i),\end{aligned}$$

where $\prod$ is the cartesian product of all the preimages. So then since f is measure-preserving

$$\begin{aligned}(f_1, ..., f_n)_\# p_{\bigotimes_{i=1}^n X_i}(y_1, ..., y_n) &= p_{\bigotimes_{i=1}^n X_i}((f_1, ..., f_n)^{-1}(y_1, ..., y_n)) \\ &= p_{\bigotimes_{i=1}^n X_i}(\prod_{i=1}^n f_i^{-1}(y_i)) \\ &= \prod_{i=1}^n p_{X_i}(f_i^{-1}(y_i)) \\ &= \prod_{i=1}^n f_{i\#} p_{X_i}(y_i) \\ &= \prod_{i=1}^n p_{Y_i}(y_i) \\ &= p_{\bigotimes_{i=1}^n Y_i}(y_1, ..., y_n),\end{aligned}$$

which completes the proof. □

Definition 3.4.2. *Let X and Y be two finite probability spaces with a measure-preserving map $f : X \to Y$ and let $n \in \mathbb{N}$ be a natural number. Then we define*

$$(f^n : X^n \to Y^n) := ((f, ..., f) : \bigotimes_{i=1}^n X \to \bigotimes_{i=1}^n Y)$$

to be the n-fold product of the measure-preserving map.

Definition 3.4.3. *Let $n \in \mathbb{N}$ be a natural number. We define $\mathbf{n} := ([n], p_{[n]})$, where $[n] = \{1, ..., n\}$ and $p_{[n]}(i) = \frac{1}{n}$ for every $i \in [n]$. We call $\mathbf{n}$ the n-point space.*

Lemma 3.4.4. *Let $n \in \mathbb{N}$ be a natural number. Then the point space is a finite probability space.*

Proof. We first note that $p_{[n]}([n]) = \frac{1}{n} \cdot n = 1$, so $p_{[n]}$ is a probability measure. Furthermore $[n]$ is finite, so $\mathbf{n}$ is a finite probability spaces. □

Proposition 3.4.5. *The 1-point space is a terminal object in* **Prob***.*

Proof. We need to show that for every finite probability space Y, there is a unique morphism $Y \to \mathbf{1}$. Let Y be a finite probabiilty space. Define $f : Y \to \mathbf{1}$ by

$$f(y) = 1.$$

Then this is a reduction as

$$f_\# p_Y(x) = p_Y(f^{-1}(x)) = p_Y(Y) = 1.$$

Moreover, this reduction is unique as if there were another reduction $g : Y \to \mathbf{1}$, then for all $y \in Y$ with $p_Y(y) > 0$,

$$g(y) = x = f(y).$$

So the one point space is a terminal object of **Prob**. □

Lemma 3.4.6. *Let $n, m \in \mathbb{N}$ be natural numbers. Let $k = n \cdot m$. Then $\mathbf{n} \otimes \mathbf{m} \simeq \mathbf{k}$.*

Proof. We first start by unraveling the definitions. Notice that $\mathbf{n} \otimes \mathbf{m} = ([n] \times [m], p_{\frac{1}{n}} \otimes p_{\frac{1}{m}})$ where for every $(i, j) \in [n] \times [m]$ we have

$$\begin{aligned} p_{[n]} \otimes p_{[m]}(i, j) = p_{[n]}(i) p_{[m]}(i) &= \frac{1}{nm} \\ &= p_{[nm]}(x) \end{aligned}$$

for all $x \in [k]$. Now we also have a bijection $f : [n] \times [m] \to [nm]$ as

$$\mathrm{card}([n] \times [m]) = \mathrm{card}([n]) \cdot \mathrm{card}([m]) = nm = \mathrm{card}([nm]) = \mathrm{card}([k]).$$

We note that such a bijection f is measure-preserving as for every $x \in [k]$, the set $f^{-1}(x)$ consists of a unique element, which implies

$$\begin{aligned} f_{\#}(p_{[n]} \otimes p_{[m]})(x) &= p_{[n]} \otimes p_{[m]}(f^{-1}(x)) \\ &= \frac{1}{nm} = p_{[k]}(x). \end{aligned}$$

Now since we have a bijective measure-preserving map between the underlying sets, by Lemma 3.2.7, $\mathbf{n} \otimes \mathbf{m} \simeq \mathbf{k}$. □

The following lemma might seem not very helpful but becomes crucial in later proofs about Grothendieck topologies being subcanonical. This lemma is a tool to obtain measure-preserving maps which switch two elements.

Lemma 3.4.7. *Suppose X is a finite probability space and let $x_1, x_2 \in X$ such that they both have positive measure. Then we have another finite probability space V with two measure-preserving maps $f, g : V \to X$ such that*

$$\begin{aligned} &f : x_1 \mapsto x_1 \text{ and } f : x_2 \mapsto x_2 \\ &g : x_1 \mapsto x_2 \text{ and } f : x_2 \mapsto x_1. \end{aligned}$$

Proof. Without loss of generality, we may assume that $p_X(x_1) \leq p_X(x_2)$. Define the space $V := (X \cup \{z\}, p_V)$ where z is not an element of X and

$$p_V(x) = \begin{cases} p_X(x_1) & \text{if } x = x_2, \\ p_X(x_2) - p_X(x_1) & \text{if } x = z, \\ p_X(x), & \text{otherwise.} \end{cases}$$

This is indeed a finite probability space as

$$\begin{aligned} p_V(V) = &\sum_{x \in X \cup \{z\}} p_V(x) \\ =&p_V(X \backslash \{x_2, z\}) + p_V(x_2) + p_V(z) \\ =&p_X(X \backslash \{x_2, z\}) + p_X(x_1) + (p_X(x_2) - p_X(x_1)) \\ =&(1 - p_X(x_2)) + p_X(x_2) = 1. \end{aligned}$$

Define the maps $f : X \cup \{z\} \to X$ and $g : X \cup \{z\} \to X$ by

$$f(x) := \begin{cases} x_2, & \text{if } x = z, \\ x, & \text{otherwise,} \end{cases}$$

and

$$g(x) := \begin{cases} x_1, & \text{if } x = x_2, \\ x_2, & \text{if } x = x_1 \text{ or } x = z, \\ x, & \text{otherwise.} \end{cases}$$

We show these maps are measure-preserving.
Let $x \in X$ such that $x \neq x_2$. Then

$$f_\# p_V(x) = p_V(f^{-1}(x)) = p_V(x) = p_X(x).$$

If $x = x_2$, we have

$$\begin{aligned} f_\# p_V(x_2) &= p_V(f^{-1}(x_2)) \\ &= p_V(\{x_2, z\}) = p_V(x_2) + p_V(z) \\ &= p_X(x_1) + (p_X(x_2) - p_X(x_1)) \\ &= p_X(x_2). \end{aligned}$$

So f is measure-preserving.
We now show g is measure-preserving, so suppose $x \in X$ such that $x \neq x_1$ and $x \neq x_2$. Then

$$g_\# p_V(x) = p_V(g^{-1}(x)) = p_V(x) = p_X(x).$$

Suppose $x = x_1$. Then

$$g_\# p_V(x_1) = p_V(g^{-1}(x_1)) = p_V(x_2) = p_X(x_1).$$

Suppose $x = x_2$. Then

$$\begin{aligned} g_\# p_V(x_2) &= p_V(g^{-1}(x_2)) = p_V(\{x_1, z\}) \\ &= p_V(x_1) + p_V(z) \\ &= p_X(x_1) + (p_X(x_2) - p_X(x_1)) \\ &= p_X(x_2). \end{aligned}$$

So f and g are measure-preserving maps and moreover

$$\begin{aligned} &f : x_1 \mapsto x_1 \text{ and } f : x_2 \mapsto x_2 \\ &g : x_1 \mapsto x_2 \text{ and } f : x_2 \mapsto x_1, \end{aligned}$$

which is what we needed to show. □

3.5 The size of Prob

Definition 3.5.1. *A category $\mathcal{C}$ is called small if* $\mathrm{Ob}(\mathcal{C})$ *is a set and for all* $X, Y \in \mathrm{Ob}(\mathcal{C})$, $\mathrm{Hom}_{\mathcal{C}}(X, Y)$ *are sets.*

We now denote by U the universe of all sets. It is worth noting that this is a notion which is different from a set.

Corollary 3.5.2. *The category **Prob** is not small.*

Proof. Suppose **Prob** is small. Then $\mathrm{Ob}(\mathbf{Prob})$ is a set. Now for every set X, $\{X\}$ is a finite set. Define the probability distribution $p_{\{X\}} : \{X\} \to [0, 1]$ by

$$p_{\{X\}}(x) = 1.$$

Then

$$(\{X\}, p_{\{X\}}) \in \mathrm{Ob}(\mathbf{Prob}),$$

from which we see

$$\{(\{X\}, p_{\{X\}}) \mid X \in U\} \subseteq \mathrm{Ob}(\mathbf{Prob}).$$

Since $\mathrm{Ob}(\mathbf{Prob})$ is a set, the union $\bigcup_{X \in U}\{X\}$ is also a set. But

$$\bigcup_{X \in U}\{X\} = U,$$

and since U is not a set, we have a contradiction. □

Definition 3.5.3. *A category $\mathcal{C}$ is called locally small if for all $X, Y \in \mathrm{Ob}(\mathcal{C})$, $\mathrm{Hom}_{\mathcal{C}}(X, Y)$ is a set.*

Proposition 3.5.4. *The category **Set** is locally small.*

Proof. Let X, Y be sets. Notice that by definition, a function $f : X \to Y$ is just a relation on $X \times Y$. So the set of all functions from X to Y, $\mathrm{Hom}_{Set}(X, Y)$, is a subset of $\mathcal{P}(X \times Y)$. Since X, Y are sets, $X \times Y$ is a set and so $\mathcal{P}(X \times Y)$ is a set. So $\mathrm{Hom}_{Set}(X, Y)$ is a set. □

Corollary 3.5.5. *The category **FinSet** is locally small.*

Proof. We note that **FinSet** is just a full subcategory of **Set**. So by Lemma 3.5.4, **FinSet** is a locally small category. □

Corollary 3.5.6. *The category **Prob** is locally small.*

Proof. Note that since a reduction is an equivalence class of measure-preserving maps we have a surjective function $f \mapsto [f]$. So we have for all finite probability spaces (X, p_X) and (Y, p_Y) an embedding

$$\mathrm{Hom}_{\mathbf{Prob}}((X, p_X), (Y, p_Y)) \hookrightarrow \{f : X \to Y \mid f \text{ is measure-preserving}\} \subseteq \mathrm{Hom}_{\mathbf{Set}}(X, Y),$$

and by Lemma 3.5.4, we have that **Prob** is locally small. □

Definition 3.5.7. *A category $\mathcal{C}$ is called essentially small if there is a small category $\mathcal{C}'$ such that $\mathcal{C} \simeq \mathcal{C}'$.*

Lemma 3.5.8. *A category $\mathcal{C}$ is essentially small if and only if it is locally small and it has a small number of isomorphism classes.*

Proof. See [8, p. 23]. □

Lemma 3.5.9. *The category **Prob** is essentially small.*

Proof. Notice that by Lemma 3.5.8 and Corollary 3.5.6, it suffices to show that **Prob** has a small number of isomorphism classes. Define for all $n \in \mathbb{N}$ the set

$$C_n := \{([n], p_n) \mid p_n \text{ is a probability measure on } [n]\}.$$

Then define the set

$$C := \bigcup_{n \in \mathbf{N}} C_n.$$

Let (X, p_X) be a finite probability space. Then Lemma 3.2.6, $(X, p_X) \simeq (\mathrm{supp}(X), p_X)$ and since $\mathrm{supp}(X)$ is fintite, we have a measure-preserving bijection $(\mathrm{supp}(X), p_X) \to ([n], p_X)$ where $([n], p_X) \in C$ and since we have a measure-preserving bijection, $(\mathrm{supp}(X), p_X) \simeq ([n], p_X)$ which proves the statement. □

4 Sheaves

In this section, we will follow the definitions and notation of [5] and [6] closely. This section serves as a reminder for what sheaves and Grothendieck topologies are for a general category $\mathcal{C}$. We first define what coverages and sieves are and can then explore Grothendieck topologies on a given category. We then formally introduce sheaves on a category that is endowed with a Grothendieck topology.

4.1 Coverages

To define Grothendieck topologies on a category, it is often useful to define coverages first, and then see what Grothendieck topology is generated by the coverage.

Definition 4.1.1. *Let $\mathcal{C}$ be a category and X an object in $\mathcal{C}$. Let H be a set of morphisms all with codomain U. We denote by*

$$\langle H \rangle := \{ f \circ g \mid f \in H,\ g \text{ a morhpism, s.t. } \mathrm{dom}(f) = \mathrm{cod}(g) \}.$$

We call $\langle A \rangle$ the sieve generated by H. If $H = \{f_1, ..., f_n\}$ is finite, we omit the brackets in notation, i.e. $\langle f_1, ..., f_n \rangle := \langle \{f_1, ..., f_n\} \rangle$.

It is worth noting that by construction, $\langle H \rangle$ is a sieve.

Lemma 4.1.2. *Let H be a collection of morphisms in a category $\mathcal{C}$ all with codomain U. Then*

$$\langle H \rangle = \bigcup_{h \in H} \langle h \rangle.$$

Proof. ($\subseteq$) Suppose $g \in \langle H \rangle$. Then $g \in \langle g \rangle \subseteq \bigcup_{h \in H} \langle h \rangle$, which implies $g \in \bigcup_{h \in H} \langle h \rangle$.
($\supseteq$) Suppose $g \in \bigcup_{h \in H} \langle h \rangle$. Then there is an $h \in H$ such that $g \in \langle h \rangle$. Then there is an f such that $g = (h \circ f) \in \langle H \rangle$. $\square$

Definition 4.1.3 ([6, p. 537])**.** *A coverage on a category $\mathcal{C}$ is a function $\mathcal{J}$ that assigns to each object U of $\mathcal{C}$ a collection $\mathcal{J}(U)$ of families of morphisms $\{f_i : U_i \to U\}_{i \in I}$, called $\mathcal{J}$-covering families of U, or if no confusion is likely, covering families of U, such that if $\mathcal{U} := \{f_i : U_i \to U\}_{i \in I}$ is a $\mathcal{J}$-covering family of an object U and $g : V \to U$ is a morphism, then there is a $\mathcal{J}$-covering family $\mathcal{V} := \{h_j : V_i \to V\}_{j \in J}$ of V such that for each $(h_j : V_j \to V) \in \mathcal{V}$, there is an $(f_i : U_i \to U) \in \mathcal{U}$ and another morphism $k : V_j \to U_i$ such that the diagram*

$$\begin{tikzcd} V_j \arrow[r, "k"] \arrow[d, "h_j"'] & U_i \arrow[d, "f_i"] \\ V \arrow[r, "g"'] & U \end{tikzcd}$$

commutes.

We will now give an important example of a coverage.

Example 4.1.4. Let X be a topological space and denote by $\mathrm{Op}(X)$ the category of open subsets of X and as morphisms the inclusions between these subsets (so we have a unique morphism $V \to U$ if and only if $V \subseteq U \subseteq X$).
Let $U \subseteq X$ be an open subset of X, so an object of $\mathrm{Op}(X)$. We can identify morphisms with codomain U with open subsets V contained in U as we have a unique morphism for every open subset of U. We can now define the function $\mathcal{J}$ such that for each object U of $\mathrm{Op}(X)$, we have that $\mathcal{J}(U)$ is the collection of all the covering families $\mathcal{U}$ of open subsets such that $\bigcup \mathcal{U} = U$.
Claim: *This is indeed a coverage.*

Proof. Let U be an open subset of X, $g : V \to U$ a morphism and $\mathcal{U}$ a covering family of U. Define $\mathcal{V} := \{V \cap W \mid W \in \mathcal{U}\}$. Then since we have a morphism $g : V \to U$, by definition, $V \subseteq U$, so $\bigcup \mathcal{V} = V \cap \bigcup \mathcal{U} = V$, which implies $\mathcal{V}$ is a covering family for V. Let $h_j : V \cap W_j \to V$ be a morphism. Then by definition, $W_j \in \mathcal{U}$ and therefore we have a morphism $f : W_j \to U$. Now we also have another morphism $k : V \cap W_j \to W_j$ as $V \cap W_j \subseteq W_j$. Then we have the commutative diagram

$$\begin{tikzcd} V \cap W_j \arrow[r, "k"] \arrow[d, "h_j"'] & W_j \arrow[d, "f"] \\ V \arrow[r, "g"'] & U \end{tikzcd}$$

which implies $\mathcal{J}$ is a coverage. □

If we have a collection of coverages, we can form a new coverage which is the pointwise union. This is a nice way of creating larger coverages from ones we already have.

Lemma 4.1.5. *Let $\{\mathcal{J}_i\}_{i \in I}$ be a collection of coverages on a category $\mathcal{C}$. Define $\mathcal{J}$ by*

$$\mathcal{J}(U) := \bigcup_{i \in I} \mathcal{J}_i(U),$$

for every object U in $\mathcal{C}$. Then $\mathcal{J}$ is a coverage.

Proof. Let $\mathcal{U}$ be a J-covering family over U and $f : V \to U$ be a morphism. Then $\mathcal{U}$ is a J_i-covering family for some $i \in I$. So we have a $\mathcal{J}_i$-covering family $\mathcal{V}_i$ over V such that for every $h_i^j \in \mathcal{V}_i$ we have a commutative diagram,

$$\begin{tikzcd} V_i^j \arrow[r, "k_j"] \arrow[d, "h_i^j"'] & U_i \arrow[d, "f_i"] \\ V \arrow[r, "f"'] & U \end{tikzcd}$$

where $f_i \in \mathcal{U}$. Then since $\mathcal{V}_i$ is a $\mathcal{J}_i$-covering family, it is by definition also a $\mathcal{J}$-covering family. So $\mathcal{J}$ is a coverage. □

4.2 Sieves

Definition 4.2.1 ([5, p. 37])**.** *Let $\mathcal{C}$ be a category and $U \in \mathrm{Ob}(\mathcal{C})$ be an object. A sieve over U, is a collection S of morphisms all with codomain U such that if $f \in S$ and for a morphism g in the category $\mathcal{C}$, $f \circ g$ is defined, then $f \circ g \in S$. We denote by*

$$\mathrm{Siev}(U) := \{S \mid S \textit{ is a sieve over } U\}$$

the collection of all sieves on U.

Example 4.2.2. Let P be a poset. We can view P as a category with objects the elements of P and we have a unique morphism $p \to q$ if and only if $p \leq q$. Then a sieve over an object p is a set S consisting of elements $q \in P$ such that $q \leq p$ and if $q \in S$, and $r \leq q \leq p$, then $r \in S$. So if an element q is in the sieve S over p, then so is any element smaller than q.

Definition 4.2.3 ([6, p. 537])**.** *A coverage is called sifted if all covering families are sieves.*

We can transform any coverage into a sifted coverage.

Proposition 4.2.4. *Let $\mathcal{J}$ be a coverage on a category $\mathcal{C}$. Then the function $\overline{\mathcal{J}}$ defined by $\overline{\mathcal{J}}(X) := \{\langle U \rangle \mid U \in \mathcal{J}(X)\}$ is a sifted coverage and it is the smallest sifted coverage where all coverings sieves of $\overline{\mathcal{J}}$ contain all coverings of $\mathcal{J}$.*

Proof. See [9, p. 17,18]. □

We will now give some examples of sieves.

Definition 4.2.5 ([5, p. 38]). *For an object U of a category $\mathcal{C}$, the collection $t_U := \{f : V \to U \mid V \in Ob(\mathcal{C})\}$ is called the maximal sieve over U.*

Lemma 4.2.6. *The maximal sieve t_U is a sieve over an object U in a category $\mathcal{C}$.*

Proof. Let $f \in t_U$. Suppose composition $f \circ g$ is defined. Then since $\mathrm{dom}(f \circ g) \in \mathrm{Ob}(\mathcal{C})$, it holds that $f \circ g \in t_U$, which implies t_U is a sieve over U. □

Lemma 4.2.7. *Let U be an object of a category $\mathcal{C}$. Then if a sieve S over U contains the identity morphism on U, it is the maximal sieve t_U.*

Proof. Let $f : V \to U$ be a morphism. Then since $\mathrm{id}_U \in S$, and since S is a sieve $f = (\mathrm{id}_U \circ f) \in S$. So any morphism with codomain U is in S, which implies $S = t_U$. □

Lemma 4.2.8 ([5, p. 38]). *Suppose U is an object of a category $\mathcal{C}$. Let S is a sieve over U and $f : V \to U$ be any morphism, then the collection*

$$f^*(S) := \{g \mid cod(g) = V, \text{ and } f \circ g \in S\},$$

is a sieve over V. We call this the pullback sieve of S along f.

Proof. Let $h \in f^*(S)$. Then by definition $f \circ h \in S$. Suppose we have a morphism g such that the composition $h \circ g$ is defined. Then since $f \circ h \in S$ and S is a sieve over U,

$$f \circ h \circ g \in S,$$

and so $h \circ g \in f^*(S)$ since we also have $\mathrm{cod}(h \circ g) = V$. □

Lemma 4.2.9. *Suppose U is an object of a category $\mathcal{C}$. Then the empty set $\emptyset$ is a sieve over U.*

Proof. Suppose the empty set is not a sieve. Notice that there are morphisms f and g such that $f \in \emptyset$ but $f \circ g \notin \emptyset$. Then this is an immediate contradiction as $f \notin \emptyset$. □

4.3 Grothendieck topologies

We now define what a Grothendieck topology is and give some examples which can be endowed on any category.

Definition 4.3.1 ([5, p. 110]). *Let $\mathcal{C}$ be a category. A Grothendieck topology on $\mathcal{C}$ is a function J which takes an object U of $\mathcal{C}$ and maps it to a collection of sieves $J(U)$ such that*

- *(maximality) the maximal sieve t_U is in $J(U)$;*
- *(stability) if $S \in J(U)$, then $f^*(S) \in J(V)$, for all $f : V \to U$;*
- *(transitivity) if $S \in J(U)$ and R is any sieve over U such that for all $h : V \to U$ in S, we have $h^*(R) \in J(V)$, then $R \in J(U)$.*

If $S \in J(U)$, we say that S J-covers U or that S is a J-covering sieve over U. If no confusion is likely, we omit J in notation, i.e. we say a sieve S covers U or S is a covering sieve over U.

We first start by giving some elementary properties of Grothendieck topologies that will be of later use.

Lemma 4.3.2. *Suppose J is a Grothendieck topology on a category $\mathcal{C}$. Then J is a coverage.*

Proof. Let S be a covering sieve, so a covering family for an object U, and $g : V \to U$ be a morphism. Choose the covering family for V, the pullback sieve of S along g, namely $g^*(S)$. Let $(h_j : V_j \to V) \in g^*(S)$. We need to choose an $(f_i : U_i \to U) \in S$ and another morphism $k : V_j \to U_i$ such that $g \circ h_j = f_i \circ k$. Choose $f_i = g \circ h_j$ which is in S since $h_j \in g^*(S)$. Then choose $k = \mathrm{id}_{V_j}$ and we obtain the commuting diagram

$$\begin{tikzcd} V_j \arrow[r, "\mathrm{id}_{V_j}"] \arrow[d, "h_j"'] & V_j \arrow[d, "g\circ h_j"] \\ V \arrow[r, "g"'] & U \end{tikzcd}$$

which implies J is a coverage. □

Lemma 4.3.3 ([5, p. 111])**.** *Let $\mathcal{C}$ be a category with a Grothendieck topology J. Suppose S is a covering sieve over an object U in $\mathcal{C}$ and let R be a sieve containing S. Then R is also a covering sieve of U.*

Proof. Let $(h : V \to U) \in S$. Then since $R \supseteq S$, we have that $h \circ \mathrm{id}_V = h \in R$, and since R is a sieve, $\mathrm{id}_V \in h^*(R)$. Then by Lemma 4.2.7, $h^*(R) = t_V$ and by maximality, t_V is a covering sieve of V. So since S is a covering sieve over U and since for every $h \in S$ we have that $h^*(R)$ is a covering sieve over V, by the transitivity axiom, R is a covering sieve over U. □

Lemma 4.3.4 ([5, p. 111])**.** *Let $\mathcal{C}$ be a category with a Grothendieck topology J. Suppose S and R are covering sieves over an object U in $\mathcal{C}$. Then $S \cap R$ is a covering sieve of U.*

Proof. Suppose $(h : V \to U) \in R$. Then

$$\begin{aligned} h^*(S \cap R) &= \{g \mid h \circ g \in S \cap R\} \\ &= \{g \mid h \circ g \in S \text{ and } h \circ g \in R\} \\ &= \{g \mid h \circ g \in S\} \cap \{g \mid h \circ g \in R\} \\ &= h^*(S) \cap t_V \\ &= h^*(S), \end{aligned}$$

and since S is a covering sieve over U, $h^*(S)$ is a covering sieve over V by the stability axiom, so $h^*(S \cap R)$ is a covering sieve over V for every $h \in R$. Then since R is a covering sieve over U, by the transitivity axiom, the sieve $S \cap R$ is a covering sieve over U. □

Lemma 4.3.5 ([9, p. 15])**.** *Let $\{J_i \mid i \in I\}$ be a collection of Grothendieck topologies on $\mathcal{C}$. Then define J to be the function that takes an object U of $\mathcal{C}$ and gives the family of covering sieves*

$$J(U) := \bigcap_{i \in I} J_i(U).$$

Then J is a Grothendieck topology.

Proof. (*maximality*): Let U be an object of $\mathcal{C}$. Then by definition, t_U is a J_i-covering sieve for all $i \in I$, so it is also a J-covering sieve.
(*stability*): Let S be a J-covering sieve over U and $f : V \to U$ be a morphism. We need to show $f^*(S)$ is a J-covering sieve over V. Since $f^*(S)$ is a J-covering sieve, it is by definition a J_i-covering sieve for every $i \in I$, and since every J_i is a Grothendieck topology, $f^*(S)$ is a J_i-covering sieve over V for every $i \in I$. So $f^*(S)$ is a J-covering sieve over V.
(*transitivity*): Suppose S is a J-covering sieve over U, and R is any sieve over U such that for every $f \in S$, the pullback $f^*(R)$ is a J-covering sieve over V. We need to show that R is a J-covering sieve over U. By definition, S is a J_i-covering sieve over U for every $i \in I$. Then for every $f \in S$, the pullback $f^*(R)$ is a J-covering sieve, so also a J_i-covering sieve for every $i \in I$. So since every J_i is a Grothendieck topology, R is a J_i-covering sieve over U for every $i \in I$. Then R is a J covering sieve over U by definition. □

Now that we know that Grothendieck topologies are closed under pointwise intersections, we can generate a Grothendieck topology from a coverage.

Definition 4.3.6. *Let $\mathcal{J}$ be a coverage. Define J to be the pointwise intersection of all Grothendieck topologies containing the coverage $\mathcal{J}$. We call J the Grothendieck topology generated by $\mathcal{J}$.*

4.4 Examples of Grothendieck topologies

Just as for topologies on sets, there are two canonical Grothendieck topologies one can define on a category, the trivial topology and the discrete topology. Both of these Grothendieck toplogies are the extremes one can choose: the trivial topology being the smallest (so the least amount of covering sieves) and the discrete topology being the largest (so the most amount of covering sieves).

Proposition 4.4.1 ([5, p. 113])**.** *Let $\mathcal{C}$ be a category. Then the function $J_{\text{triv}} : U \mapsto \{t_U\}$ is a Grothendieck topology on $\mathcal{C}$. We call J_{triv} the trivial topology.*

Proof. (*maximality*): Let U be an object of $\mathcal{C}$. Then by definition, the maximal sieve $t_U \in J_{\text{triv}}(U) = \{t_U\}$.
(*stability*): Let $f : V \to U$ be a morphism. Then

$$\begin{aligned} f^*(t_U) &= \{g \mid \text{cod}(g) = V, \text{ and } h \circ g \in t_U\} \\ &= \{g : W \to V \mid W \in \text{Ob}(\mathcal{C})\} \\ &= t_V \in J_{\text{triv}}(V). \end{aligned}$$

It is worth noting that this is a weaker notion than a Grothendieck topology as every Grothendieck topology, is also a coverage which we will formulate in the next lemma.
transitivity): Suppose R is a sieve over U such that for all $h : V \to U$ in t_U, we have $h^*(R) \in J_{\text{triv}}(V)$. Then we have that

$$h^*(R) = t_V.$$

Then id_U induces

$$t_U = \text{id}_U^*(R) = \{g \mid \text{cod}(g) = U, \text{ and } g \in R\} = R.$$

So R is a covering sieve. □

Proposition 4.4.2. *Let $\mathcal{C}$ be a category. Then the function J_d that assigns to every object U the collection of all sieves over U is a Grothendieck topology on $\mathcal{C}$. We call J_d the discrete topology.*

Proof. (*maximality*): Let U be an object of $\mathcal{C}$. Then $t_U \in J_d(U)$ as t_U is a sieve.
(*stability*): Let U be an object of $\mathcal{C}$, $S \in J_d(U)$ and $f : V \to U$ be a morphism. Then $f^*(S) \in J_d(V)$ as $f^*(S)$ is a sieve.
(*transitivity*): Let U be an object of $\mathcal{C}$. Let $S \in J_d(U)$ and R be any sieve over U such that for every $f \in S$, we have $f^*(R) \in J_d(V)$. Then $R \in J_d(U)$ as R is a sieve. □

We now give another Grothendieck topology one can endow on any category. At a later stage, we will analyze this Grothendieck topology on **Prob**.

Proposition 4.4.3 ([5, p. 115])**.** *Let J_{de} be the function such that*

$$U \mapsto J_{de}(U) := \{S \textit{ sieve over } U \mid \forall f : V \to U, \exists g : W \to V, \textit{ such that } f \circ g \in S\}.$$

Then J_{de} is a Grothendieck topology on $\mathcal{C}$. We call J_{de} the dense topology.

Proof. (*maximality*): Let U be an object of $\mathcal{C}$. We show that the maximal sieve, t_U, is in $J_{de}(U)$. Let $f : V \to U$ be any morphism. Choose $g = \mathrm{id}_V$. Then

$$f \circ \mathrm{id}_V = (f : V \to U) \in t_U$$

and since t_U is a sieve, $t_U \in J_{de}(U)$.
(*stability*): Let $S \in J_{de}(U)$ be a covering sieve over U. Let $f : V \to U$ be a morphism in $\mathcal{C}$ We need to show that $f^*(S) \in J_{de}(V)$. We have the following equivalences:

$$\begin{aligned} f^*(S) \in J_{de}(V) &\iff \forall g : V \to U,\ \exists h : W \to V, \text{ such that } g \circ h \in f^*(S), \\ &\iff \forall g : V \to U,\ \exists h : W \to V, \text{ such that } f \circ (g \circ h) \in S. \end{aligned}$$

Now since $S \in J_{de}(U)$, we have

$$\forall g' : V' \to U,\ \exists h' : W' \to V' \text{ such that } g' \circ h' \in S,$$

So since $f \circ g : V \to U$, there is an $h' : W \to V$ such that $f \circ g \circ h \in S$. So choosing $h = h'$ gives us

$$f \circ g \circ h \in S.$$

So $f^*(S) \in J_{de}(V)$.
(*transitivity*): Suppose $S \in J_{de}(U)$ and suppose that R is a sieve over U such that for all morphisms $h : V \to U$ in S we have $h^*(R) \in J_{de}(V)$. We need to show that

$$R \in J_{de}(U) \iff \forall g : V \to U,\ \exists f : W \to V \text{ such that } g \circ f \in R.$$

Let $h \in S$. Since $h^*(R) \in J_{de}(V)$, we have

$$\begin{aligned} h^*(R) \in J_{de}(A) &\iff \forall g' : V' \to U,\ \exists f' : W' \to V', \text{ such that } g' \circ f' \in h^*(R), \\ &\iff \forall g' : V' \to U,\ \exists f : W' \to V', \text{ such that } h \circ (g' \circ f') \in R. \end{aligned}$$

So id_U induces

$$\mathrm{id}_U^*(R) \iff \forall g' : V' \to U,\ \exists f : W' \to U', \text{ such that } \mathrm{id}_U \circ (g' \circ f') = g' \circ f' \in R.$$

Let $g : W \to U$. Then we can simply choose again $f = f'$. So then we have

$$g \circ f \in R,$$

so $R \in J_{de}(U)$. Therefore, the dense topology J_{de} is a Grothendieck topology. $\square$

In a lot of categories, the dense topology has a different characterization which is easier to work with. We will now work on such a characterization.

Definition 4.4.4. *A category $\mathcal{C}$ is said to satisfy the (right) Ore condition if for all for diagrams,*

$$\begin{tikzcd} & X \arrow[d, "f"] \\ Y \arrow[r, "g"'] & Z \end{tikzcd}$$

there exists another object C in $\mathcal{C}$ with two morphisms, $k : C \to X$, and $l : C \to Y$, such that the extended diagram,

$$\begin{tikzcd} C \arrow[r, "k"] \arrow[d, "l"'] & X \arrow[d, "f"] \\ Y \arrow[r, "g"'] & Z \end{tikzcd}$$

commutes.

Definition 4.4.5. *Let $\mathcal{C}$ be a category satisfying the right Ore condition. Let J_{at} be the function such that*

$$U \mapsto J_{at}(U) = \{S \text{ sieve over } U \mid S \neq \emptyset\}.$$

We call J_{at} the atomic topology.

Proposition 4.4.6. *Let $\mathcal{C}$ be a category satisfying the Ore condition. Then J_{at} is a Grothendieck topology on $\mathcal{C}$.*

Proof. (*maximality*): Notice that for all objects $U \in \text{Ob}(\mathcal{C})$, the maximal sieve $t_U \in J_{\text{at}}(U)$ as $\text{id}_X \in t_U$ which implies it is nonempty.
(*stability*): Suppose $S \in J_{\text{at}}(U)$ and let $f : V \to U$ be any morphism in $\mathcal{C}$. Then since S is nonempty, there is a morphism $(g : W \to U) \in S$. So we have a diagram

$$\begin{tikzcd} & V \arrow[d, "f"] \\ W \arrow[r, "g"'] & U \end{tikzcd}$$

which, by the Ore condition, can be completed to the commutatitive diagram

$$\begin{tikzcd} X \arrow[r, "k"] \arrow[d, "l"'] & V \arrow[d, "f"] \\ W \arrow[r, "g"'] & U. \end{tikzcd}$$

So we have a morhpism $k : X \to V$ such that

$$f \circ k = g \circ l,$$

and since $g \in S$, we have $f \circ k = g \circ l \in S$, and therefore $k \in f^*(S)$. So $f^*(S)$ is nonempty which implies $f^*(S) \in J_{\text{at}}(V)$.
(transitivity): Suppose $S \in J_{\text{at}}(U)$ and suppose R is a sieve over U such that for all $h : V \to U$ in S, we have $h^*(R) \in J_{\text{at}}(V)$. We need to show $R \in J_{\text{at}}(U)$ i.e. R is nonempty. Since $S \in J_{\text{at}}(U)$, it is nonempty and therefore we have a morphism $(h : V \to U) \in S$. Then $h^*(R) \in J_{\text{at}}(V)$, which implies it is nonempty as well and we can choose a morphism $(g : W \to V) \in h^*(R)$ which implies, by definition, $h \circ g \in R$ and therefore R is nonempty.
So J_{at} is a Grothendieck topology. □

Proposition 4.4.7. *Let $\mathcal{C}$ be a category satisfying the Ore condition. Then the atomic topology and the Dense topology are the same,*

$$J_{at} = J_d.$$

Proof. We need to show that for every object U in $\mathcal{C}$,

$$J_{\text{at}}(U) = J_d(U).$$

We first show $J_d(U) \subseteq J_{\text{at}}(U)$. Suppose $S \in J_d(U)$. Then for the identity id_U, by definition of the dense topology, there is a morhpism $g : V \to U$ such that $\text{id}_U \circ g = g \in S$ which implies S is nonempty and therefore, $S \in J_{\text{at}}(U)$.
We now show $J_{\text{at}}(U) \subseteq J_d(U)$. Let $S \in J_{\text{at}}(U)$ and let $f : V \to U$ be a morphism. Since S is nonempty, we have a morphism $(g : W \to U) \in S$. So we have a diagram

$$\begin{tikzcd} & V \arrow[d, "f"] \\ W \arrow[r, "g"'] & U \end{tikzcd}$$

which, by the Ore condition, can be completed to the commutatitive diagram

$$\begin{tikzcd} X \arrow[r, "k"] \arrow[d, "l"'] & V \arrow[d, "f"] \\ W \arrow[r, "g"'] & U. \end{tikzcd}$$

So we have a morhpism $k : X \to V$ such that

$$f \circ k = g \circ l,$$

Now since $g \in S$ and S is a sieve, $f \circ k = g \circ l \in S$, so $S \in J_d(U)$. □

4.5 Presheaves

Definition 4.5.1 ([5, p. 25]). *A presheaf F on a category $\mathcal{C}$ is a functor*

$$F : \mathcal{C}^{op} \to \textbf{\textit{Set}}.$$

It is worth noting that a presheaf F is just an object of the category $[\mathcal{C}^{\text{op}}, \textbf{Set}]$. We then define

$$\text{Psh}(\mathcal{C}) = [\mathcal{C}^{\text{op}}, \textbf{Set}]$$

as the category of presheaves on $\mathcal{C}$. Recall that for two objects of $\text{Psh}(\mathcal{C})$ (so two presheaves), F and G, the morphisms between them are the natural transformations between F and G.

Definition 4.5.2. *We call a presheaf F on a category $\mathcal{C}$ a representable presheaf if it is of the form $Hom_{\mathcal{C}}(-, U)$ for some object U in $\mathcal{C}$.*

Example 4.5.3. Recall that for all objects Z of $\mathcal{C}$, we have the Yoneda Functor $\mathbf{y}^Z : \mathcal{C}^{\text{op}} \to \textbf{Set}$, which is defined by

$$\mathbf{y}^Z(U) = \text{Hom}_{\mathcal{C}}(U, Z)$$

for all objects U in $\mathcal{C}$, and for all morphisms $f : V \to U$ in $\mathcal{C}$, we get a map

$$\begin{aligned} \mathbf{y}^Z(f) &: \text{Hom}_{\mathcal{C}}(U, Z) \to \text{Hom}_{\mathcal{C}}(V, Z), \\ &: g \mapsto g \circ f. \end{aligned}$$

This is a presheaf as $\mathbf{y}^Z \in \text{Psh}(\mathcal{C})$.

Definition 4.5.4. *Let I be a set. The functor $\Delta_I : \mathcal{C}^{op} \to \textbf{\textit{Set}}$ defined by*

$$\begin{aligned} \Delta_I(U) &= I \\ \Delta_I(f) &= id_I \end{aligned}$$

for all objects U and morphisms f in $\mathcal{C}$ is called the constant presheaf.

Definition 4.5.5 ([5, p. 36]). *Let $\mathcal{C}$ be a locally small category. A subfunctor of a presheaf F on $\mathcal{C}$, is another presheaf G on $\mathcal{C}$ such that for all objects U in $\mathcal{C}$ and each morphism $f : U \to V$ in $\mathcal{C}$,*

- *$G(U) \subseteq F(U)$;*
- *$G(f) : G(V) \to G(U)$ is a restriction of $F(f) : F(V) \to F(U)$, i.e.*

$$F(f)|_{G(V)} = G(f).$$

If G is such a subfunctor of F, we denote it by $G \subset F$. We denote by

$$\text{SubFct}(F) = \{G \in \text{Psh}(\mathcal{C}) \mid G \subset F\}$$

the collection of all subfunctors of F.

The next lemma, which is stated in [5, p. 37], allows us to transform a sieve over an object Z uniquely into a subfunctor of the representable presheaf $\mathbf{y}^Z$ and vice versa.

Lemma 4.5.6. *Let $\mathcal{C}$ a locally small category and Z an object in $\mathcal{C}$. Then there is a bijection between the set of all subfunctors of the Yoneda Functor $\boldsymbol{y}^Z$ and the collection of all sieves on Z, i.e.*

$$\mathrm{SubFct}(\boldsymbol{y}^Z) \simeq \mathrm{Siev}(Z).$$

Proof. We define $\phi : \mathrm{SubFct}(\mathbf{y}^Z) \to \mathrm{Siev}(Z)$ by

$$\phi(G) = \{f \mid \exists U \in \mathrm{Ob}(\mathcal{C}), \text{ such that } f \in G(U)\}.$$

Let $G \in \mathrm{SubFct}(\mathbf{y}^Z)$. We first check $\phi(G)$ a sieve over Z. Let $f \in \phi(G)$. Since $G \subset \mathbf{y}^Z$, we have $G(U) \subseteq \mathrm{Hom}_{\mathcal{C}}(U, Z)$, so f is a morphism from U to Z and therefore, $\phi(G)$ is a collection of morphisms with codomain Z. Suppose we have a morphism $g : V \to U$ such that the composition $f \circ g : V \to Z$ exists. We need to show $f \circ g \in \phi(G)$, so it suffices to show that $f \circ g \in G(V)$. Now,

$$\begin{aligned}
f \circ g &= \mathrm{Hom}_{\mathcal{C}}(g, Z)(f) \\
&= \mathbf{y}^Z(g)(f) \\
&= \mathbf{y}^Z(g)|_{G(U)}(f) && \text{(since } f \in G(U)) \\
&= G(g)(f). && \text{(since } G \subset \mathbf{y}^Z)
\end{aligned}$$

We notice that since $g : V \to U$,

$$G(g) : G(U) \to G(V)$$

This implies that

$$f \circ g = G(g)(f) \in G(V).$$

So $f \circ g \in \phi(G)$, so $\phi(G)$ is a sieve.
We now define $\psi : \mathrm{Siev}(Z) \to \mathrm{SubFct}(\mathbf{y}^Z)$, by

$$\psi(S) = G_S : \mathcal{C}^{\mathrm{op}} \to \mathbf{Set},$$

which is the functor defined by, for all objects U in $\mathcal{C}$ and all morphisms $f : Z \to Y$,

$$G_S(U) := S \cap \mathrm{Hom}_{\mathcal{C}}(U, Z) \subseteq \mathrm{Hom}_{\mathcal{C}}(U, Z), \tag{3}$$
$$G_S(g) := \mathbf{y}^Z(g). \tag{4}$$

Then this is clearly a subfunctor of $\mathbf{y}^Z$ as for each morphism $g : Y \to Z$ and for each $f \in G_S(Z)$, we have

$$G_S(g)(f) = \mathbf{y}^Z(g)(f) = \mathbf{y}^Z(g)|_{G(Z)}(f).$$

We now want to show that for all $S \in \mathrm{Siev}(Z)$ and for all $G \in \mathrm{SubFct}(\mathbf{y}^Z)$

$$(\phi \circ \psi)(S) = S$$
$$(\psi \circ \phi)(G) = G.$$

We have

$$\begin{aligned}
(\phi \circ \psi)(S) &= \phi(G_S) \\
&= \{f \mid \exists U \in \mathrm{Ob}(\mathcal{C}), f \in G_S(U)\} \\
&= \{f \mid \exists U \in \mathrm{Ob}(\mathcal{C}), f \in S \cap \mathrm{Hom}_{\mathcal{C}}(U, Z)\} \\
&= S.
\end{aligned}$$

Now we also have for all objects V in $\mathcal{C}$,

$$\begin{aligned}(\psi \circ \phi)(G)(V) &= \psi(\{f \mid \exists U \in \mathrm{Ob}(\mathcal{C}),\ f \in G(U)\})(V) \\ &= \{f \mid \exists U \in \mathrm{Ob}(\mathcal{C}),\ f \in G(U)\} \cap \mathrm{Hom}_{\mathcal{C}}(V, Z) \\ &= \{f \mid f \in G(V)\} \cap \mathrm{Hom}_{\mathcal{C}}(V, Z) \\ &= G(V),\end{aligned}$$

and for all $g : U \to V$, and all $h \in G(V)$

$$\begin{aligned}(\psi \circ \phi)(G)(g)(h) &= \psi(\{f \mid \exists U \in \mathrm{Ob}(\mathcal{C}),\ f \in G(U)\})(g)(h) \\ &= \mathbf{y}^Z(g)(h) = G(g)(h)\end{aligned}$$

where we used that $G \subset \mathbf{y}^Z$. Then we have that

$$\begin{aligned}\phi \circ \psi &= \mathrm{id}_{\mathrm{Siev}(Z)} \\ \psi \circ \phi &= \mathrm{id}_{\mathrm{SubFct}(\mathbf{y}^Z)}\end{aligned}$$

So we find

$$\mathrm{SubFct}(\mathbf{y}^Z) \simeq \mathrm{Siev}(Z)$$

which is what we needed to show. □

4.6 Sheaves

We will now define what sheaves are on a Grothendieck topology. There is a way of defining sheaves over the weaker notion of coverages, but we will omit this way as we do not need it in this report.

Definition 4.6.1 ([5, p. 122]). *Let $\mathcal{C}$ be a category and J a Grothendieck topology on $\mathcal{C}$. Suppose S is a J-cover of an object U in $\mathcal{C}$. Suppose $F : \mathcal{C}^{\mathrm{op}} \to$ **Set** is a presheaf. A matching family for S of F is a natural transformation*

$$x : G_S \to F \tag{5}$$

where G_S is the subfunctor via the identification in Lemma 4.5.6. We denote such a matching family by $(x_f)_{f \in S}$.

There is a characterization of matching families which is easier to work with.

Proposition 4.6.2 ([5, p. 121]). *A matching family $(x_f)_{f \in S}$ for a sieve S over U of a presheaf F is the same as a rule that assigns a morphism $f : V \to U$ in S to an element $x_f \in F(V)$ such that for every morphism $g : W \to V$.*

$$F(g)(x_f) = x_{f \circ g}.$$

Proof. By Lemma 4.5.6, given a sieve S over U, we can choose the unique subfunctor as defined in 3,

$$G_S \subset \mathbf{y}^U.$$

Then a natural transformation $x : G_S \to F$ consists for every morphism $g : W \to V$, of a diagram

$$\begin{tikzcd} G_S(W) \arrow[r, "x_W"] \arrow[d, "G_S(g)"'] & F(W) \arrow[d, "F(g)"] \\ G_S(V) \arrow[r, "x_V"'] & F(V) \end{tikzcd}$$

which commutes. Now since G_S is a subfunctor,

$$G_S(V) \subseteq \mathrm{Hom}_{\mathcal{C}}(V, U) \text{ and } G_S(W) \subseteq \mathrm{Hom}_{\mathcal{C}}(W, U). \tag{6}$$

Also, G_S being a subfunctor implies that

$$G_S(g) \text{ is a restriction of } \mathbf{y}^U(g) : h \mapsto h \circ g.$$

Define for all $f : W \to U$ and $h : V \to U$

$$\begin{aligned} x_f :=& x_W(f), \\ x_h :=& x_V(h). \end{aligned}$$

Then chasing a morphism $(f : W \to U) \in G_S(W)$ around gives

$$(F(g) \circ x_W)(f) = F(g)(x_f) = (x_V \circ G_S)(g)(f) = x_V(f \circ g) = x_{f \circ g}.$$

□

Example 4.6.3. Let $\mathcal{C}$ be a category and endow it with the trivial topology J_{triv}. Let U be an object of $\mathcal{C}$ which is J_{triv}-covered by S. Then S is the maximal sieve t_U by definition of the trivial topology. Let $f : A \to U$ be a morphism in t_U. We would like to find a matching family of t_U for the Yoneda Functor $\mathbf{y}^U : \mathcal{C}^{\text{op}} \to \mathbf{Set}$. Then by Proposition 4.6.2, a matching family for t_U of $\mathbf{y}^U$ is then a rule such that for every morphism $f \in t_U$,

$$(f : V \to U) \mapsto x_f \in \mathbf{y}^U(V) = \mathrm{Hom}_{\mathcal{C}}(V, U)$$

such that for every other morphism $g : W \to V$,

$$\mathbf{y}^U(g)(x_f) = x_{f \circ g}.$$

Assign to every morphism f, the same morphism f. Then $(x_f)_{f \in S}$ is clearly a matching family for t_U of S as,

$$\mathbf{y}^U(g)(x_f) = x_f \circ g = f \circ g = x_{f \circ g}. \tag{7}$$

Definition 4.6.4 ([5, p. 121])**.** *Let $\mathcal{C}$ be a category and J be a Grothendieck topology on $\mathcal{C}$. Suppose S is a J-cover of an object U in $\mathcal{C}$. Suppose $F : C^{op} \to$ **Set** is a presheaf. Suppose we have a matching family $(x_f)_{f \in S}$ for S of F. An amalgamation of such a matching family is a single element $x \in F(U)$ such that*

$$F(f)(x) = x_f,$$

for all $f \in S$.

Example 4.6.5. Recall Example 4.6.3. We would like to see what an amalgamation is of the matching familiy $(x_f)_{f \in S}$. An amalgamation is an element

$$x \in \mathbf{y}^U(U) = \mathrm{Hom}_{\mathcal{C}}(U, U)$$

such that for all $(f : V \to U) \in t_U$

$$\mathbf{y}^U(f)(x) = x_f.$$

We choose the element $x = \mathrm{id}_U$. Then

$$\mathbf{y}^U(f)(x) = x \circ f = \mathrm{id}_U \circ f = f = x_f.$$

So x_{id_X} is an amalgamation of the matching family $(x_f)_{f \in S}$.

Definition 4.6.6 ([5, p. 122]). *Suppose F is a presheaf on a category $\mathcal{C}$ and let J be a Grothendieck topology on $\mathcal{C}$. Then F is called a J-sheaf, or if no confusion is likely a sheaf, if for every object U, every J-cover S of U and every matching family $(x_f)_{f\in S}$ of S, there is a unique amalgamation for $(x_f)_{f\in S}$.*

Notice that all sheaves are still presheaves. So a morphism of sheaves is just a morphism of presheaves which is a natural transformation.

Definition 4.6.7 ([5, p. 128]). *Let $\mathcal{C}$ be a category and J be Grothendieck topology on $\mathcal{C}$. We define $\mathrm{Sh}_J(\mathcal{C})$ to be the category of sheaves with as objects the J-sheaves and morphisms the natural transformations between two J-sheaves.*

Proposition 4.6.8. *Let $\mathcal{C}$ be a category. Consider the trivial topology J_{triv} as defined in Definition 4.4.1. Then any presheaf F is a sheaf.*

Proof. Let $F : \mathcal{C}^{\mathrm{op}} \to \mathbf{Set}$ be any presheaf on $\mathcal{C}$. Let $X \in \mathrm{Ob}(\mathcal{C})$. Let S be a T-cover of X, so $S = t_X$. Let $(x_f)_{f\in S}$ be a matching family for t_X of F. So we have for all morphisms $f : A \to X$, and for all morphisms $g : B \to A$,

$$F(g)(x_f) = x_{f\circ g}.$$

Then since t_X is the maximal sieve, it contains the identity which induces for all $g : B \to X$,

$$F(g)(x_{\mathrm{id}_X}) = x_{\mathrm{id}_X\circ g} = x_g.$$

This implies that x_{id_X} is an amalgamation for the matching family $(x_f)_{f\in S}$. We are left to show uniqueness. Suppose there is another $x \in F(X)$ such that for all $g : B \to X$

$$F(g)(x) = x_g.$$

Then, since F is a functor and x is assumed to be an amalgamation,

$$x = \mathrm{id}_{F(X)}(x) = F(\mathrm{id}_X)(x) = x_{\mathrm{id}_X}.$$

So $x = x_{\mathrm{id}_X}$ which implies that x_{id_X} is unique. So the presheaf F is a sheaf. □

Proposition 4.6.9. *Let $\mathcal{C}$ be a category and J a Grothendieck topology on $\mathcal{C}$. Then the constant presheaf $\Delta_{\{pt\}} : \mathcal{C}^{op} \to$ **Set**, is a sheaf. Here, $\{pt\}$ is the set with one point.*

Proof. Let U be an object of $\mathcal{C}$, let S be a covering sieve of C and $(x_f)_{f\in S}$ a matching family for S of $\Delta_{\{\mathrm{pt}\}}$. Then we have for all morphisms $f : V \to U$ and all morphisms $g : W \to V$,

$$\Delta_{\{\mathrm{pt}\}}(g)(x_f) = x_{f\circ g}.$$

Now notice by definition we have $\Delta_{\{\mathrm{pt}\}}(g)(x_f) = \mathrm{id}_{\{\mathrm{pt}\}}(x_f) = x_f = \mathrm{pt}$ since $x_f \in \{\mathrm{pt}\}$. We want to choose an amalgamation of this matching family, so an element in $\Delta_{\{\mathrm{pt}\}}(U) = \{\mathrm{pt}\}$ and we are left with one choice, namely $x = \mathrm{pt}$. So by our previous argument, for every $f : V \to U$, we have

$$\Delta_{\{\mathrm{pt}\}}(f)(x) = \mathrm{pt} = x_f,$$

so x is an amalgamation. This amalgamation is moreover unique, since there is no other element in $\{\mathrm{pt}\}$. Therefore, $\Delta_{\{\mathrm{pt}\}}$ is a sheaf. □

Definition 4.6.10 ([5, p. 126]). *A Grothendieck topology J on a category $\mathcal{C}$ is called subcanonical if for every object U in $\mathcal{C}$, its representable presheaf $Hom_{\mathcal{C}}(-, U)$ is a J-sheaf.*

By the Yoneda Lemma (see [5, p. 26], all subcanonical Grothendieck topologies allow a full and faithfull embedding

$$\mathcal{C} \xhookrightarrow{\mathbf{y}} \mathrm{Sh}_J(\mathcal{C}),$$

which is the Yoneda embedding. This allows us to enlarge a category $\mathcal{C}$ harmlessly to $\mathrm{Sh}_J(\mathcal{C})$, as long as J is subcanonical.

Lemma 4.6.11. *Let J and K be two Grothendieck topologies on a category $\mathcal{C}$ such that every J-covering sieve is also a K-covering sieve. Then every K-sheaf is also a J-sheaf. In other words $\mathrm{Sh}_K(\mathcal{C}) \subseteq \mathrm{Sh}_J(\mathcal{C})$.*

Proof. Suppose that F is a K-sheaf. Then for every object U, every K-covering sieve S over U, every matching family of F over S has a unique amalgamation. But since every J-covering sieve is also a K covering sieve, we have that F must also be a J-sheaf. □

With Lemma 4.6.11, it makes sense for us to wonder what the largest Grothendieck topology J on a category $\mathcal{C}$ is, such that J is subcanonical.

Definition 4.6.12 ([5, p. 126])**.** *Let $\mathcal{C}$ be a category. The largest Grothendieck topology J that is subcanonical is called the canonical topology on $\mathcal{C}$.*

In general, a description of the canonical topology on a category $\mathcal{C}$ is hard to find. On **Prob**, it turns out that the canonical topology is a well-know Grothendieck topology which we will explore at a later stage.

5 Grothendieck topologies on Prob

In this section, we give a large class of examples of Grothendieck topologies on **Prob** using the theory of coverages guided by [6, Chapter C2]. To see that these are new examples of Grothendieck topologies on **Prob**, we show that they are different from the known topologies, so different from J_{triv}, J_d, J_{at}.

5.1 The X-projection topology

Proposition 5.1.1. *Consider **Prob** and any finite probability space X. Then for every finite probability space U we assign the $\mathcal{J}_X$-covering family $\mathcal{U} := \{\pi_U : U \otimes X \to U\}$. Then $\mathcal{J}_X$ is a coverage on **Prob**. We call this the X-projection coverage.*

Proof. Let $\{\pi_U : U \otimes X \to U\}$ be a covering family of U. Let V be a finite probability space and $g : V \to U$ be a reduction. Choose the covering family $\{\pi_V : V \otimes X \to V\}$. Let $h \in \{\pi_V : V \otimes X \to V\}$. Then $h = \pi_V$. Choose $k = (g, \mathrm{id}_X)$. Then indeed, the diagram

$$\begin{tikzcd} V \otimes X \arrow[r, "(g,\mathrm{id}_X)"] \arrow[d, "\pi_V"'] & U \otimes X \arrow[d, "\pi_U"] \\ V \arrow[r, "g"'] & U \end{tikzcd}$$

commutes, as

$$\pi_U \circ (g, \mathrm{id}_X) = g \circ \pi_V.$$

$\square$

It is worth noting that the $\mathcal{J}_X$-coverages can be transformed into a sifted coverage $\overline{\mathcal{J}_X}$ by Proposition4.2.4, where for each object U of **Prob**, the covering sieves over U are only the sieves generated by the projections, i.e.

$$\overline{\mathcal{J}_X}(U) = \{\langle \pi_U \rangle\}$$

Note that this is in general not a Grothendieck topology on **Prob**.
We would now like to investigate what Grothendieck topology is generated by the $\mathcal{J}_X$-coverage.

Lemma 5.1.2. *Let J be a Grothendieck topology on **Prob** such that for all U in **Prob**, $\overline{\mathcal{J}_X}(U) \subseteq J(U)$. Then any sieve S over U containing a projection $\pi_U^n : X^n \otimes U \to U$ for some $n \in \mathbb{N}$ is a covering sieve over U in the Grothendieck topology J.*

Proof. We prove this by induction.
Suppose $\pi_U^1 \in S$. Then since S is a sieve, any $\pi_U^1 \circ f \in S$, and therefore $\langle \pi_U^1 \rangle \subseteq S$. Then by Lemma 4.3.3 and since $\langle \pi_U^1 \rangle$ is covering, S is covering.
Let $k \in \mathbb{N}$. Suppose all sieves S containing a projection $\pi_U^k : X^k \otimes U \to U$ are covering. We want to show that all sieves containing $\pi_U^{k+1} : X^{k+1} \otimes U \to U$ are covering. Since $\langle \pi_U^1 \rangle$ is a covering sieve over U, by the transitivity axiom, it suffices to show that for every $(g : V \to U) \in \langle \pi_U^1 \rangle$, we have for all sieves R containing the morphism π_U^{k+1}, that $g^*(R)$ is a covering sieve. So let $(g : V \to U) \in \langle \pi_U^1 \rangle$. We will first show that $g^*(\langle \pi_U^{k+1} \rangle)$ is a covering sieve. By definition of the generating sieve, there is another morphism $f : V \to X \otimes U$ such that $g = \pi_U^1 \circ f$. Notice that we have

$$\begin{aligned} g^*(\langle \pi_U^{k+1} \rangle) &= (\pi_U^1 \circ f)^*(\langle \pi_U^{k+1} \rangle) \\ &= \{h \mid \pi_U^1 \circ f \circ h \in \langle \pi_U^{k+1} \rangle\}. \end{aligned}$$

Now we see that we have a commutative diagram

$$
\begin{tikzcd}
X^k \otimes V \arrow[r, "(\mathrm{id}_{X^k}, f)"] \arrow[d, "\pi_V^k"'] & X^k \otimes (X \otimes U) \arrow[d, "="] \\
V \arrow[d, "f"'] & X^{k+1} \otimes U \arrow[d, "\pi_U^{k+1}"] \\
X \otimes U \arrow[r, "\pi_U^1"'] & U
\end{tikzcd}
$$

So we see that

$$(\pi_U^1 \circ f \circ \pi_V^k) = ((\mathrm{id}_{X^k}, f) \circ \pi_U^{k+1}) \in \langle \pi_U^{k+1} \rangle.$$

Then by definition of g and the pullback sieve, we have that $\pi_U^k \in g^*(\langle \pi_U^{k+1} \rangle)$. This implies that $g^*(\langle \pi_U^{k+1} \rangle)$ is a covering sieve over V for every $g \in \langle \pi_U^1 \rangle$ which is a covering sieve over U. So by the transitivity axiom, $\langle \pi_U^{m+1} \rangle$ is also a covering sieve over U. Then if R is a sieve on U containing π_U^{m+1}, since it is a sieve $\langle \pi_U^{m+1} \rangle \subseteq R$ and then again by Lemma 4.3.3, R is also a covering sieve over U. □

Definition 5.1.3. *Let X be a finite probability space in **Prob**. We define J_X to be the function that takes an object U in **Prob** and yields a family of covering sieves of U such that a sieve S is covering over U if and only if there exists $n \in \mathbb{N}$ such that $(\pi_U^n : X^n \otimes U \to U) \in S$, where $\pi_U^n : X^n \otimes U \to U$ is the usual projection. We call this the X-projection topology and denote it by J_X.*

Theorem 5.1.4. *For all objects X in **Prob** the X-projection topology is a Grothendieck topology.*

Proof. (*maximality*): We need to show that the maximal sieve t_U is a covering sieve, so we need to find an $n \in \mathbb{N}$ such that the projection $\pi_U^n \in t_U$. Choose $n = 1$. Then since the maximal sieve contains every morphism with codomain U, $\pi_U^1 \in t_U$. So t_U is a covering sieve of U.
(*stability*): Let S be a covering sieve over U and let $f : V \to U$ be a morphism. Then there is an $n \in \mathbb{N}$ such that $\pi_U^n \in S$. We need to show that $f^*(S)$ is a covering sieve of V, so we need to show that there is a $k \in \mathbb{N}$ such that $\pi_V^k \in f^*(S)$. Note that we have the commutative square

$$
\begin{tikzcd}
X^n \otimes V \arrow[r, "(\mathrm{id}_{X^n}, f)"] \arrow[d, "\pi_V^n"'] & X^n \otimes U \arrow[d, "\pi_U^n"] \\
V \arrow[r, "f"'] & U
\end{tikzcd}
$$

so we choose $k = n$ and we have by commutativity that $f \circ \pi_V^k \in S$ which by definition of the pullback sieve implies that $\pi_V^k \in f^*(S)$.
(*transitivity*): Let S be a covering sieve of U and let R be another sieve on U such that for every $(f : V \to U) \in S$, we have that $f^*(R)$ is a covering sieve of V. We need to show that there exists an $n \in \mathbb{N}$ such that $\pi_U^n \in R$. Since S is a covering sieve of U, there is a $k \in \mathbb{N}$ such that the projection $\pi_U^k \in S$. Now by assumption, there is an $m \in \mathbb{N}$ such that

$$\pi_{X^k \otimes U}^m \in (\pi_U^k)^*(R).$$

Then by definition of the pullback sieve,

$$\pi_U^k \circ \pi_{X^k \otimes U}^m \in R.$$

Now we also have a commutative square

$$
\begin{tikzcd}
X^m \otimes (X^k \otimes U) \arrow[r, "="] \arrow[d, "\pi_{X^k \otimes U}^m"'] & X^{m+k} \otimes U \arrow[d, "\pi_U^{m+k}"] \\
X^k \otimes U \arrow[r, "\pi_U^k"'] & U
\end{tikzcd}
$$

where the isomorphism is canonical. Therefore $\pi^k_U \circ \pi^m_{X^k \otimes U} = \pi^{k+m}_U \in R$. □

Proposition 5.1.5. *The X-projection topology is the Grothendieck topology generated by the $\mathcal{J}_X$-coverage.*

Proof. Suppose there is a Grothendieck topology smaller than the X-projection topology that contains the $\mathcal{J}_X$-coverage. Then there is a sieve S on an object U which is covering in the X-projection topology but not covering in the smaller Grothendieck topology. But since S is a covering sieve over U in the X-projection topology, there is an $n \in \mathbb{N}$ such that $\pi^n_U \in S$, but by Lemma 5.1.2 then this is also a covering sieve in the Grothendieck topology generated by the $\mathcal{J}_X$ coverage, which is a contradiction. □

Proposition 5.1.6. *The **1**-projection topology is the same as the trivial topology.*

Proof. Let U be an object of **Prob**. Then we see that for every $n \in \mathbb{N}$, $\mathbf{1}^n \otimes U \simeq U$ by Lemma 3.4.6. So for every $n \in \mathbb{N}$, the projection $\pi^n_U = \mathrm{id}_U$. So a sieve is covering if and only if it contains the identity, and therefore every covering sieve of U is the maximal sieve on U, which is exactly the trivial topology. □

Proposition 5.1.7. *For all objects X in **Prob**, such that there are two different points $x, y \in X$ with $p_X(x) > 0$ and $p_X(y) > 0$, the X-projection topology is different from the trivial topology.*

Proof. To show that the X-projection topology is not the trivial topology, we need to find a covering sieve on an object U that is not the maximal sieve t_U. Let $U = \mathbf{1}$. Notice that we have the morphism $\pi^1_{\mathbf{1}} : X \to \mathbf{1}$. Let $S = \langle \pi^1_{\mathbf{1}} \rangle$. Then since X has two points of positive measure, there is no morphism $\mathbf{1} \to X$, which in particular means that there is no morphism $f : \mathbf{1} \to X$ such that $\pi^1_{\mathbf{1}} \circ f = \mathrm{id}_{\mathbf{1}}$. Then this implies that $\langle \pi^1_{\mathbf{1}} \rangle \neq t_{\mathbf{1}}$ as $\mathrm{id}_{\mathbf{1}} \notin \langle \pi^1_{\mathbf{1}} \rangle$ so the X-projection topology is not the trivial topology as they have different covering sieves. □

Proposition 5.1.8. *For all objects X in **Prob**, the X-projection topology is different from the atomic topology. (see Proposition 4.4.6)*

Proof. We need to show that there is a sieve S on an object U that is not a covering sieve in the X-projection topology, so that there is a nonempty sieve S on an object such that it does not contain any of the projections $\pi^n_U : X^n \otimes U \to U$. Let $U = \mathbf{1}$. Define for all $n \in \mathbb{N}$, the set

$$S_n = \{p_{X^n}(A) \mid A \subset X^n\} \subseteq [0,1]$$

Then for all $n \in \mathbb{N}$, the set S_n is finite as there are only finitely many subsets of X^n as X^n is also finite. So in particular, for all $n \in \mathbb{N}$, the set S_n is countable. Then since the countable union of countable sets is countable, the set

$$S = \bigcup_{n \in \mathbb{N}} S_n \subseteq [0,1]$$

is also countable, since $\mathbb{N}$ is countable.
Suppose we have a morphism $f : X^n \to Y$ for some $n \in \mathbb{N}$. Then for all $y \in Y$,

$$p_Y(y) = p_{X^n}(f^{-1}(y)) \in S_n,$$

as $f^{-1}(y) \subseteq X^n$. Now since S is countable, we can choose a $q \in [0,1]$ such that $q \notin S$ and $(1-q) \notin S$. Then define $Q = (\{0,1\}, p)$, where $p(0) = q$ and $p(1) = 1-q$. Now notice that for all $n \in \mathbb{N}$, there is no morphism $f : X^n \to Q$, as if there were one,

$$q = p(0) = p_{X^n}(f^{-1}(0)) \in S^n \subseteq S,$$

which is a contradiction since $q \notin S$. Now we do have a morphism $g : Q \to \mathbf{1}$, which is just the constant function. Then the sieve generated by g, namely

$$\langle g \rangle := \{g \circ h \mid g : A \to Q \text{ such that } A \in \mathrm{Ob}(\mathbf{Prob})\},$$

is not a covering sieve, as if it were, there would exist an $n \in \mathbb{N}$ such that $(\pi_{\mathbf{1}}^n : X^n \to 1) \in \langle g \rangle$, which by definition of the sieve would imply that there is a morphism $f : X^n \to Q$ which we have shown does not exist. So we have found a nonempty sieve which is not a covering sieve which implies that the X-projection topology is different from the atomic topology. $\square$

Proposition 5.1.9. *For all objects X in **Prob**, the X-projection topology is different from the discrete topology.*

Proof. Notice that a sieve is covering in the X-projection topology if and only if it contains a projection, so in particular, all covering sieves are nonempty in the X-projection topology. However, in the discrete topology, the empty sieves are also covering sieves. So the discrete topology and the projection topology are different on **Prob**. $\square$

5.2 The projection topology

Proposition 5.2.1. *Consider **Prob**. Then for every finite probability space U we assign the $\mathcal{J}_{proj}$-covering family $\mathcal{U} := \{\pi_U : U^2 \to U\}$. Then $\mathcal{J}_{proj}$ is a coverage on **Prob** and we call it the projection coverage.*

Proof. Let $\{\pi_U : U^2 \to U\}$ be a covering family of U. Let V be another finite probability space and $f : V \to U$ be a reduction. Choose the covering family $\{\pi_V : V^2 \to V\}$. Let $h \in \{\pi_V : V^2 \to V\}$. Then $h = \pi_V$. Choose $k = f^2$. Then indeed, the diagram

$$\begin{tikzcd} V^2 \arrow[r, "f^2"] \arrow[d, "\pi_V"'] & U^2 \arrow[d, "\pi_U"] \\ V \arrow[r, "f"'] & U \end{tikzcd}$$

commutes, as for every $(v_1, v_2) \in V^2$ with positive measure,

$$\begin{aligned}(\pi_U \circ f^2)(v_1, v_2) &= \pi_U(f(v_1), f(v_2)) \\ &= f(v_1) = (f \circ \pi_V)(v_1, v_2).\end{aligned}$$

So $\mathcal{J}_{\mathrm{proj}}$ is a coverage. $\square$

We can again turn this into a sifted coverage by Proposition 4.2.4 and then see what Grothendieck topology it generates. Again, we would like a characterization for this Grothendieck topology.

Lemma 5.2.2. *Let J be a Grothendieck topology on **Prob** such that for all U in **Prob**, $\overline{\mathcal{J}_{proj}}(U) \subseteq J(U)$. Then any sieve S over U containing a projection $\pi_U^n : U^{n+1} \to U$ for some $n \in \mathbb{N}$ is a covering sieve over U in the Grothendieck topology J.*

Proof. We prove this again by induction.
Suppose $\pi_U^1 \in S$. Then since S is a sieve, any $\pi_U^1 \circ f \in S$, and therefore $\langle \pi_U^1 \rangle \subseteq S$. Then by Lemma 4.3.3 and since $\langle \pi_U^1 \rangle$ is covering, S is covering.
Let $k \in \mathbb{N}$. Suppose any sieve S containing a projection $\pi_U^k : U^{k+1} \to U$ is covering. We want to show that any sieve containing $\pi_U^{k+1} : U^{k+2} \to U$ is covering. Since $\langle \pi_U^1 \rangle$ is a covering sieve, by the transitivity axiom, it suffices to show that for every $(g : V \to U) \in \langle \pi_U^1 \rangle$, we have for all sieves R containing the morphism π_U^{k+1}, that $g^*(R)$ is a covering sieve as R then is also a covering sieve. So let $(g : V \to U) \in \langle \pi_U^1 \rangle$, then by definition, there is another morphism $f : V \to U^2$ such that $g = \pi_U^1 \circ f$. Notice that

$$\begin{aligned}g^*(\langle \pi_U^{k+1} \rangle) &= (\pi_U^1 \circ f)^*(\langle \pi_U^{k+1} \rangle) \\ &= \{h \mid \pi_U^1 \circ f \circ h \in \langle \pi_U^{k+1} \rangle\}.\end{aligned}$$

We also have a commutative diagram

$$\begin{tikzcd}
V^{k+1} \arrow[r, "f^{k+1}"] \arrow[d, "\pi_V^k"'] & U^{2(k+1)} \arrow[d, "\pi^k_{U^{k+2}}"] \\
V \arrow[d, "f"'] & U^{k+2} \arrow[d, "\pi_U^{k+1}"] \\
U^2 \arrow[r, "\pi_U^1"'] & U
\end{tikzcd}$$

So we see that

$$(g \circ \pi_V^k) = (\pi_U^1 \circ f \circ \pi_V^k) = (\pi_U^{k+1} \circ \pi^k_{U^{k+2}} \circ f^{k+1}) \in \langle \pi_U^{k+1} \rangle$$

therefore we have that $\pi_V^k \in g^*(\langle \pi_U^{k+1} \rangle)$. This implies that $g^*(\langle \pi_U^{k+1} \rangle)$ is a covering sieve over V for every $g \in \langle \pi_U^1 \rangle$ which is a covering sieve over U. So by the transitivity axiom, $\langle \pi_U^{k+1} \rangle$ is also a covering sieve over U. Then if R is a sieve on U containing π_U^{k+1}, since it is a sieve $\langle \pi_U^{k+1} \rangle \subseteq R$ and then again by Lemma 4.3.3, R is also a covering sieve over U. □

Definition 5.2.3. *We define J_{proj} to be the function that takes an object U in **Prob** and yields a family of covering sieves of U such that a sieve S is covering over U if and only if there exists $n \in \mathbb{N}$ such that $(\pi_U^n : U^{n+1} \to U) \in S$, where $\pi_U^n : U^{n+1} \to U$ is the usual projection. We call this the projection topology and denote it by J_{proj}.*

Theorem 5.2.4. *The projection topology is a Grothendieck topology on **Prob**.*

Proof. (*maximality*): We need to show that the maximal sieve t_U is a covering sieve, so we need to find an $n \in \mathbb{N}$ such that the projection $\pi_U^n \in t_U$. Choose $n = 1$. Then since the maximal sieve contains every morphism with codomain U, $\pi_U^1 \in t_U$. So t_U is a covering sieve of U.
(*stability*): Let S be a covering sieve of U and let $f : V \to U$ be a morphism. Then there is an $n \in \mathbb{N}$ such that $\pi_U^n \in S$. We need to show that $f^*(S)$ is a covering sieve of V, so we need to show that there is an $k \in \mathbb{N}$ such that $\pi_V^k \in f^*(S)$. Note that we have the commutative square

$$\begin{tikzcd}
V^{n+1} \arrow[r, "f^{n+1}"] \arrow[d, "\pi_V^n"'] & U^{n+1} \arrow[d, "\pi_U^n"] \\
V \arrow[r, "f"'] & U
\end{tikzcd}$$

so we choose $k = n$ and we have by commutativity that $f \circ \pi_V^k \in S$ which by definition of the pullback sieve implies that $\pi_V^k \in f^*(S)$.
(*transitivity*): Let S be a covering sieve of U and let R be another sieve on U such that for every $(f : V \to U) \in S$, we have that $f^*(R)$ is a covering sieve of V. We need to show that there exists an $n \in \mathbb{N}$ such that $\pi_U^n \in R$. Since S is a covering sieve of U, there is a $k \in \mathbb{N}$ such that the projection $\pi_U^k \in S$. Now by assumption, there is an $m \in \mathbb{N}$ such that

$$\pi^m_{X^k \otimes U} \in (\pi_U^k)^*(R).$$

Then by definition of the pullback sieve,

$$\pi_U^k \circ \pi^m_{U^{k+1}} \in R.$$

Now we also have a commutative diagram

$$\begin{tikzcd}
U^{k+m+1} \arrow[d, "\pi^m_{U^{k+1}}"'] \arrow[dr, "\pi_U^{m+k}"] & \\
U^{k+1} \arrow[r, "\pi_U^k"'] & U
\end{tikzcd}$$

Therefore $\pi_U^k \circ \pi_{U^{k+1}}^m = \pi_U^{k+m} \in R$, which implies R is covering. □

Proposition 5.2.5. *The projection topology is the Grothendieck topology generated by the projection coverage.*

Proof. Suppose there is a Grothendieck topology smaller than the projection topology that contains the projection coverage. Then there is a sieve S on an object U which is covering in the projection topology but not covering in the smaller Grothendieck topology. But since S is a covering sieve over U in the projection topology, there is an $n \in \mathbb{N}$ such that $\pi_U^n \in S$, but by Lemma 5.2.2 then this is also a covering sieve in the Grothendieck topology generated by the projection coverage, which is a contradiction. □

Lemma 5.2.6. *The only projection topology covering sieve of the one point space is the maximal sieve.*

Proof. Notice that on the one point space $\mathbf{1}$ a sieve is covering if and only if it contains a projection $\pi_{\mathbf{1}}^n : \mathbf{1}^{n+1} \to \mathbf{1}$, and since $\mathbf{1}^{n+1} \simeq \mathbf{1}$, the morphism $\pi_{\mathbf{1}}^{n+1} = \mathrm{id}_{\mathbf{1}}$, since $\mathbf{1}$ has only one point. So then a sieve is covering over $\mathbf{1}$ if and only if it contains the identity which implies a sieve is covering over $\mathbf{1}$ if and only if it is the maximal sieve. □

Proposition 5.2.7. *The projection topology is different from the trivial topology.*

Proof. Consider the finite probability space $\mathbf{2}$. Then the sieve $\langle \pi_{\mathbf{2}}^1 : \mathbf{2}^2 \to \mathbf{2} \rangle$ is a covering sieve in the projection topology. It suffices to show this sieve is not the maximal sieve. Notice that $\mathbf{2}^2 \simeq \mathbf{4}$, and that there is no morphism $f : \mathbf{4} \to \mathbf{2}$. So, in particular, there is no morphism $f : \mathbf{4} \to \mathbf{2}$ such that $\pi_{\mathbf{2}}^1 \circ f = \mathrm{id}_{\mathbf{2}}$ so $\langle \pi_{\mathbf{2}}^1 \rangle$ does not contain the identity which implies $\langle \pi_{\mathbf{2}}^1 \rangle$ is not the maximal sieve. So the projection topology is different from the trivial topology. □

Proposition 5.2.8. *The projection topology is different from the X-projection topology for all X in the category* **Prob**.

Proof. By Proposition 5.1.6, the $\mathbf{1}$-projection topology is the trivial topology and by Proposition 5.2.7, the projection topology is different from the trivial topology, so the projection topology is not the $\mathbf{1}$-projection topology. Now by Lemma 5.2.6, the only covering sieve of the one point space is the maximal sieve, and in the proof of Proposition 5.1.7, the sieve $\langle \pi_{\mathbf{1}}^1 : X \to \mathbf{1} \rangle$ is also a covering sieve in the X-projection topology where X has two points of positive measure. □

Proposition 5.2.9. *The projection topology is different from the atomic topology.*

Proof. Let X be a finite probability space with two points that have strictly positive measure. Now the atomic topology declares the sieve $\langle \pi_{\mathbf{1}}^1 : X \to \mathbf{1} \rangle$ to be a covering sieve over $\mathbf{1}$, (see Proposition 5.1.7). However, this sieve is not a covering sieve in the projection topology, since the only sieve covering over the one point space is the maximal sieve by Lemma 5.2.6. So the discrete topology is different from the projection topology. □

Proposition 5.2.10. *The projection topology is different from the discrete topology on* ***Prob***.

Proof. Notice that a sieve is covering in the projection topology if and only if it contains a projection, so in particular, all covering sieves are nonempty in the projection topology. However, in the discrete topology, the empty sieves are also covering sieves. So the discrete topology and the projection topology are different on **Prob**. □

5.3 Collection projection topologies

Now that we have a large class of coverages and Grothendieck topologies, we can combine them to form larger coverages and Grothendieck topologies.

Proposition 5.3.1. *Let $\{X_i\}_{i \in I}$ be a (small) nonempty collection of objects in* ***Prob***. *Then we assign for every finite probability space U the $\mathcal{J}_{\{X_i\}_{i\in I}}$-covering family $\mathcal{U} := \{\pi_U : U \otimes X_i \to U \mid i \in I\}$. Then $\mathcal{J}_{\{X_i\}_{i\in I}}$ is a coverage on* ***Prob***.

Proof. Notice that $\mathcal{J}_{\{X_i\}_{i\in I}}(U) = \bigcup_{i\in I} \mathcal{J}_{X_i}(U)$ for every object U in **Prob**. So applying Lemma 4.1.5, we see that $\mathcal{J}_{\{X_i\}_{i\in I}}$ is a coverage. $\square$

Definition 5.3.2. *Let $\{X_i\}_{i\in I}$ be a collection of finite probability spaces. Let $K \subseteq I$ be finite and $\sigma : K \to \mathbb{N}$ be a map. Then we define the finite probability space $X^{K,\sigma}$ by*

$$X^{K,\sigma} := \bigotimes_{k\in K} X_k^{\sigma(k)}$$

and for every finite probability space U, the projection

$$\pi_U^{K,\sigma} : X^{K,\sigma} \otimes U \to U$$

as the usual projection onto U.

Lemma 5.3.3. *Let $\{X_i\}_{i\in I}$ be a nonempty collection of finite probability spaces. Let J be a Grothendieck topology on **Prob** such that for all objects U in **Prob**, $\overline{\mathcal{J}_{\{X_i\}_{i\in I}}}(U) \subseteq J(U)$. Then all sieves S over U containing a projection*

$$\pi_U^{K,\sigma} : X^{K,\sigma} \otimes U \to U$$

where $K \subseteq I$ finite and $\sigma : K \to \mathbb{N}$, is a covering sieve.

Proof. The proof of this lemma will follow the same structure as the proof of Lemma 5.1.2 and 5.2.2.

We first note that since for every $i \in I$, and every object U in **Prob** we have the chain of inclusions $\overline{\mathcal{J}_{X_i}}(U) \subseteq \overline{\mathcal{J}_{\{X_i\}_{i\in I}}}(U) \subseteq J(U)$ and then by Lemma 5.1.2 we see that every sieve S over U containing a projection $\pi_U^{n_i} : X_i^{n_i} \otimes U \to U$, where $i \in I$ and $n_i \in \mathbb{N}$, is covering. Let $K \subseteq I$ finite. We will show this by induction on K.

Suppose K consists of only one element k and suppose $\pi^{K,\sigma} \in S$ for some map $\sigma : K \to \mathbb{N}$. Then $\sigma : \{k\} \to \mathbb{N}$ so we have $\pi_U^{K,\sigma} = \pi_U^{\{k\},\sigma} = \pi_U^{\sigma(k)} \in S$ and by our previous reasoning, we have that S is covering over U.

Let $L \subseteq K$ and let $k \in K$. Suppose all sieves containing a projection $\pi_U^{L,\tau}$, for some $\tau : L \to \mathbb{N}$, are covering. Let $\rho : L \cup \{x\} \to \mathbb{N}$ be a map and R be a sieve over U containing the projection $\pi_U^{L\cup\{k\},\rho}$. We want to show R is a covering sieve over U, but we will first show $\langle \pi_U^{L\cup\{k\},\rho} \rangle$ is a covering sieve over U. Notice that $\langle \pi_U^{\rho(k)} : X_k^{\rho(k)} \otimes U \to U \rangle$ is a covering sieve over U. Let

$$(g : V \to U) \in \langle \pi_U^{\rho(k)} \rangle,$$

and by definition of the generating sieve, $g = \pi_U^{\rho(k)} \circ f$ for some $f : V \to X_k^{\rho(k)} \otimes U$. Now we have the commuting diagram

$$\begin{tikzcd}
X^{L,\rho|_L} \otimes V \arrow[r, "(\mathrm{id}, f)"] \arrow[d, "\pi_V^{L,\rho|_L}"'] & X^{L,\rho|_L} \otimes (X_k^{\rho(k)} \otimes U) \arrow[d, "\sim"] \\
V \arrow[d, "f"'] & X^{L\cup\{k\},\rho} \otimes U \arrow[d, "\pi_U^{L\cup\{k\},\rho}"] \\
X_k^{\rho(k)} \otimes U \arrow[r, "\pi_U^{\rho(k)}"'] & U
\end{tikzcd}$$

and we get that

$$\begin{aligned}
(g \circ \pi_V^{L,\rho|_L}) &= (\pi_U^{\rho(k)} \circ f \circ \pi_V^{L,\rho|_L}) \\
&= (\pi_U^{L\cup\{k\},\rho} \circ (\mathrm{id}, f)) \in \langle \pi_U^{L\cup\{k\},\rho} \rangle
\end{aligned}$$

So we see that for every $g \in \langle \pi_U^{\rho(k)} \rangle$ which is a covering sieve over U, that $\pi_V^{L,\rho|_L} \in g^*(\langle \pi_U^{L\cup\{k\},\rho} \rangle)$ by definition of the pullback sieve, and then by assumption, we have that $g^*(\langle \pi_U^{L\cup\{k\},\rho} \rangle)$ is a covering sieve over V. Then by the transitivity axiom of a Grothendieck topology, we have that $\langle \pi_U^{L\cup\{k\},\rho} \rangle$ is a covering sieve over U. Then since R contains the projection $\pi_U^{L\cup\{k\},\rho}$ and since it is a sieve, $\langle \pi_U^{L\cup\{k\},\rho} \rangle \subseteq R$. And then by Lemma 4.3.3, we have that R is also a covering sieve over U. ☐

Definition 5.3.4. *Let $\{X_i\}_{i\in I}$ be a nonempty collection of finite probability spaces. We define the function J_I to be the function that takes an object U in **Prob** and yields a family of covering sieves on U such that a sieve S is covering over U if and only if there exists a finite subcollection $K \subseteq I$ of finite probability spaces $\{X_k\}_{k\in k}$ together with a map $\sigma : K \to \mathbb{N}$, such that*

$$(\pi_U^{K,\sigma} : X^{K,\sigma} \otimes U \to U) \in S.$$

We call J_I the collection projection topology with respect to $\{X_i\}_{i\in I}$.

Theorem 5.3.5. *The collection projection topology with respect to $\{X_i\}_{i\in I}$ is a Grothendieck topology.*

Proof. (*maximality*): Notice that for t_U contains all morphisms with codomain U, so it also contains all projections.
(*stability*): Suppose S is a covering sieve over U and $f : V \to U$ is a morphism. Then S contains a projection $(\pi_U^{K,\sigma} : X^{K,\sigma} \otimes U \to U)$ for some finite $K \subseteq I$ and some map $\sigma : K \to \mathbb{N}$. We need to show that $f^*(S)$ is a covering sieve of V, so we need to show that there is a subcollection $L \subseteq I$ together with a map $\tau : L \to \mathbb{N}$ such that $\pi_V^{L,\tau} \in f^*(S)$. Choose $L := K$ and $\tau := \sigma$ Then we have the commutative diagram

$$\begin{tikzcd} X^{K,\sigma} \otimes V \arrow[r, "(\mathrm{id}_{X^{K,\sigma}}, f)"] \arrow[d, "\pi_V^{K,\sigma}"'] & X^{K,\sigma} \otimes U \arrow[d, "\pi_U^{K,\sigma}"] \\ V \arrow[r, "f"'] & U \end{tikzcd}$$

which implies

$$\pi_U^{K,\sigma} \circ (\mathrm{id}_{X^{K,\sigma}}, f) = f \circ \pi_V^{K,\sigma} \in S$$

and therefore $\pi_V^{K,\sigma} \in f^*(S)$, which implies $f^*(S)$ is covering over V.
(*transitivity*): Let S be a covering sieve over U and R be a sieve such that for every $f \in S$, the pullback $f^*(R)$ is a covering sieve. Then since S is covering, it contains a projection $\pi_U^{K,\sigma}$ for some $K \subseteq I$ finite and some map $\sigma : K \to \mathbb{N}$. Then $(\pi_U^{K,\sigma})^*(R)$ is covering over $X^{K,\sigma} \otimes U$ by definition, so $(\pi_U^K)^*(R)$ contains a projection $\pi_{X^{K,\sigma}\otimes U}^{L,\tau}$ where $L \subseteq I$ finite and $\tau : K \to \mathbb{N}$. Define $T := K \cup L$, which is also a finite subcollection of I, and define $\rho : T \to \mathbb{N}$ defined by

$$\rho(t) := \begin{cases} \sigma(t) + \tau(t), & \text{if } t \in K \cap L \\ \sigma(t), & \text{if } t \in K \backslash L \\ \tau(t), & \text{if } t \in L \backslash K. \end{cases}$$

Then we have

$$\begin{aligned} X^{L,\tau} \otimes (X^{K,\sigma} \otimes U) &= (\bigotimes_{l\in L} X_l^{\tau(l)}) \otimes (\bigotimes_{k\in K} X_k^{\sigma(k)}) \otimes U \\ &\simeq (\bigotimes_{t\in K\cap L} X_t^{\sigma(t)+\tau(t)}) \otimes (\bigotimes_{t\in K\backslash L} X_t^{\sigma(t)}) \otimes (\bigotimes_{t\in L\backslash K} X_t^{\tau(t)}) \otimes U \\ &\simeq (\bigotimes_{t\in T} X_t^{\rho(t)}) \otimes U = X^{T,\rho} \otimes U, \end{aligned}$$

from which we get a commutative diagram,

$$\begin{tikzcd} X^{L,\tau} \otimes (X^{K,\sigma} \otimes U) \arrow[r, "\sim"] \arrow[d, "\pi^{L,\tau}_{X^{K,\sigma}\otimes U}"'] & X^{T,\rho} \otimes U \arrow[d, "\pi^{T,\rho}_U"] \\ X^{K,\sigma} \otimes U \arrow[r, "\pi^{K,\sigma}_U"'] & U \end{tikzcd}$$

So we have that, since $\pi^{L}_{X^{K,\sigma}\otimes U} \in (\pi^{K,\sigma}_U)^*(R)$,

$$\pi^{K,\sigma}_U \circ \pi^{L,\tau}_{X^{K,\sigma}\otimes U} = \pi^{T,\rho}_U \in R$$

So R is a covering sieve. □

Lemma 5.3.6. *Let $\{X_i\}_{i\in I}$ be a nonempty collection of finite probability spaces. Then J_I is the Grothendieck topology generated by the $\mathcal{J}_{\{X_i\}_{i\in I}}$ coverage.*

Proof. Let J be a Grotendieck topology smaller than then J_I such that it also contains the sifted $\overline{\mathcal{J}_{\{X_i\}_{i\in I}}}$ coverage. Then there is a J-covering sieve that is not a J_I-covering sieve. But by Lemma 5.3.3, since J contains $\overline{\mathcal{J}_{\{X_i\}_{i\in I}}}$, a J-covering sieve over U contains then a projection $\pi^{K,\sigma}_U$ for some $K \subseteq I$ finite and some $\sigma : K \to \mathbb{N}$ a map, however it is then also a J-covering sieve which is a contradiction. So therefore, J_I is the Grothendieck topology generated by $\mathcal{J}_{\{X_i\}_{i\in I}}$. □

Lemma 5.3.7. *Let $\{X_i\}_{i\in I}$ be a collection of finite probability spaces. Then for every $i \in I$, and every U an object of **Prob**, we have that $J_{X_i}(U) \subseteq J_I(U)$.*

Proof. Let $i \in I$ and U an object in **Prob**. Then a J_{X_i}-covering sieve S over U contains a projection π^n_U for some $n \in N$. Then choose $\{i\} \subseteq I$ as finite subcollection together with $\sigma(i) := n$. Then $\pi^{\{i\},\sigma}_U = \pi^n_U \in S$ so S is also a J_I covering sieve over U. □

Proposition 5.3.8. *Let $\{X_i\}_{i\in I}$ be a nonempty collection of finite probability spaces. Then if there is an $i \in I$ such that X_i has two elements with positive measure, then J_I is different from J_{triv}.*

Proof. By Lemma 5.3.7, we have for all finite probability spaces U, that $J_{X_i}(U) \subseteq J(U)$ and by Proposition 5.1.7, we have that there is a J_{X_i}-covering sieve that is not a J_{triv}-covering sieve which proves the statement. □

Proposition 5.3.9. *Let $\{X_i\}_{i\in I}$ be a nonempty, finite collection of finite probability spaces. Then J_I is different from J_{at} on **Prob**. (see Proposition 4.4.6)*

Proof. We need to show that there is a nonempty sieve S on an object U that is not a J_I-covering sieve, so that there is a nonempty sieve S on an object such that it does not contain any of the projections $\pi^{K,\sigma}_U : X^{K,\sigma} \otimes U \to U$ for some $K \subseteq I$ together with some function $\sigma : K \to \mathbb{N}$. Let $U = \mathbf{1}$. Define for every $K \subseteq I$ and for every map $\sigma : K \to \mathbb{N}$, the set

$$S^{\sigma}_K := \{p_{X^{K,\sigma}}(A) \mid A \subseteq X^{K,\sigma}\} \subseteq [0,1],$$

which is a finite subset of the unit interval, as $X^{K,\sigma}$ is finite and then there are only finitely many subsets A of $X^{K,\sigma}$. Denote by F_K the collection of all maps $\sigma : K \to \mathbb{N}$, which is a countable set since K is finite. Then define

$$S_K := \bigcup_{\sigma\in F_K} S^{\sigma}_K$$

which is a countable subset of the unit interval since a countable union of countable sets is countable. Now define

$$S := \bigcup_{K\subseteq I} S_K$$

which is again countable since I is finite. Now if we have a morphism $f : X^{K,\sigma} \to Y$ for some $K \subseteq I$ finite and $\sigma : K \to \mathbb{N}$, then for all $y \in Y$,

$$p_Y(y) = p_{X^{K,\sigma}}(f^{-1}(y)) \in S_K^\sigma,$$

as $f^{-1}(y) \subseteq X^{K,\sigma}$. Now since S is a countable subset of the unit interval, we may choose a $q \in [0,1]$ such that $q \notin S$ and $1 - q \notin S$. Define $Q := (\{0,1\}, p)$ where $p(0) = q$ and $p(1) = 1 - q$. Now notice that for all $K \subseteq I$ and all maps $\sigma : K \to \mathbb{N}$, there is no morphism $f : X^{K,\sigma} \to Q$, as if there were one

$$q = p(0) = p_{X^{K,\sigma}}(f^{-1}(0)) \in S$$

which is a contradiction since $q \notin S$. Now we do have a morphism $g : Q \to \mathbf{1}$, which is just the constant function. Then the sieve generated by g, namely $\langle g \rangle$ is not a covering sieve as if it were, there would be a projection $\pi_{\mathbf{1}}^{K,\sigma} \in \langle g \rangle$, which would imply that there is a morphism $f : X^{K,\sigma} \to Q$ which we have just shown does not exist. So we have found a nonempty sieve which is not a covering sieve which implies that J_I is different from J_{triv}. □

Lemma 5.3.10. *Let $\{X_i\}_{i \in I}$ be a nonempty, finite collection of finite probability spaces. Then J_I is different from J_d on **Prob**.*

Proof. Notice that all J_I covering sieves contains a projection, and it is therefore nonempty. However, empty sieves are covering in J_d, so J_I is not J_d. □

6 The atomic topology on Prob

We now return to a familiar example of a Grothendieck topology: the atomic topology J_{at}. This Grothendieck topology can only be defined on a category if it satisfies the right Ore condition and we show that **Prob** does indeed satisfy this condition. To conclude this paper, we will prove that the atomic topology on **Prob** is subcanonical and from this powerful result, it follows that the entire class of Grothendieck topologies on **Prob** we have previously given are subcanonical.

6.1 Adhesion

Definition 6.1.1. *Let U, V and W be finite probability spaces with reductions $f : V \to U$ and $g : W \to U$ as visualised in*

$$\begin{tikzcd} & V \arrow[d, "f"] \\ W \arrow[r, "g"'] & U. \end{tikzcd}$$

We define the adhesion of U along f and g by

$$V \otimes_U W := (V \times_U W, p_{V \otimes_U W}),$$

where

$$V \times_U W := \{(v, w) \in V \times W \mid f(v) = g(w)\}$$

and for every $(v, w) \in V \times_U W$,

$$p_{V \otimes_U W}(v, w) := \frac{p_V(v) p_W(w)}{p_{V \times W}(V \times_U W)}. \tag{8}$$

Lemma 6.1.2. *The adhesion of U along f and g, $V \otimes_U W$, is a finite probability space.*

Proof. We note that $V \times_U W \subseteq V \times W$ which has finite support, so $V \times_U W$ also has finite support. Also

$$p_{V \otimes_U W}(V \times_U W) := \frac{p_{V \times W}(V \times_U W)}{p_{V \times Y}(V \times_U W)} = 1,$$

so $p_{V \otimes_U W}$ is a probability measure. □

Lemma 6.1.3. *The projections $\pi_V : V \times_Z Y \to V$ and $\pi_Y : V \times_Z Y \to Y$ are measure preserving.*

Proof. Suppose $A \subseteq V$. Then

$$\begin{aligned} \pi_V^{-1}(A) &= \{(x, y) \in V \times_U W \mid \pi_W(x, y) = x \in A\} \\ &= A \times_U W. \end{aligned}$$

So

$$\begin{aligned} (\pi_V)_\# p_{V \times_U W}(A) = p_{V \times_U W}(\pi_V^{-1}(A)) &= p_{V \times_Z W}(A \times_U W) \\ &= \frac{p_{V \times W}(A \times_U W)}{p_{V \times W}(V \times_U W)} \\ &= p_V(A). \end{aligned}$$

The proof for π_W is completely analogous. □

Proposition 6.1.4. *The category **Prob** satisfies the Ore condition.*

Proof. Let U, V and W be finite probability spaces with reductions $f : V \to U$ and $g : W \to U$. We need to find a finite probability space X with two reductions $k : X \to V$ and $l : X \to W$ such that ,

$$\begin{tikzcd} X \arrow[r, "k"] \arrow[d, "l"'] & V \arrow[d, "f"] \\ W \arrow[r, "g"'] & U \end{tikzcd}$$

commutes. We define $X = V \otimes_U W$ as the adhesion of U along f and g and the morphisms $l := \pi_V$ and $k := \pi_W$. Then for each $(x, y) \in V \times_U W$, with positive measure

$$(f \circ \pi_W)(x, y) = f(y) = g(x) = (g \circ \pi_V)(x, y). \tag{9}$$

So the diagram

$$\begin{tikzcd} V \otimes_U W \arrow[r, "\pi_X"] \arrow[d, "\pi_Y"'] & V \arrow[d, "f"] \\ W \arrow[r, "g"'] & U \end{tikzcd}$$

commutes which implies **Prob** satisfies the Ore condition. □

6.2 The atomic topology on Prob

Now that we know that **Prob** satisfies the Right Ore condition, we can endow it with the atomic topology.

Theorem 6.2.1. *The atomic topology on **Prob** is subcanonical.*

Proof. To show J_{at} is subcanonical, we need to show that for all objects Z in **Prob**, that $\mathbf{y}^Z := \mathrm{Hom}_{\mathbf{Prob}}(-, Z)$ is a sheaf. By III.4 Lemma 2 of [5], $\mathrm{Hom}_{\mathbf{Prob}}(-, Z)$ is a sheaf for the atomic topology if and only if for any $f : V \to U$ and any $\alpha \in \mathrm{Hom}_{\mathbf{Prob}}(V, Z)$, if for all commutative diagrams

$$\begin{tikzcd} W \arrow[r, shift left, "g"] \arrow[r, shift right, "h"'] & V \arrow[r, "f"] & U \end{tikzcd}$$

we have $\mathrm{Hom}_{\mathbf{Prob}}(g, Z)(\alpha) = \mathrm{Hom}_{\mathbf{Prob}}(h, Z)(\alpha)$, then there is a unique $\gamma \in \mathrm{Hom}_{\mathbf{Prob}}(U, Z)$ such that $\alpha = \mathrm{Hom}_{\mathbf{Prob}}(f, Z)(\gamma)$. By the definition of $\mathrm{Hom}_{\mathbf{Prob}}(-, Z)$ it suffices to show that for all $f : V \to U$ and all $\alpha : V \to Z$, if for all $g, h : W \to V$ such that $f \circ g = f \circ h$ we have $\alpha \circ g = \alpha \circ h$, then there is a unique $\gamma : U \to Z$ such that the diagram

$$\begin{tikzcd} W \arrow[r, shift left, "g"] \arrow[r, shift right, "h"'] & V \arrow[r, "f"] \arrow[d, "\alpha"] & U \arrow[dl, dotted, "\gamma"] \\ & Z & \end{tikzcd}$$

commutes. So let $f : V \to U$ and $\alpha : V \to Z$. Assume that for all $g, h : W \to V$ such that $f \circ g = f \circ h$ we have $\alpha \circ g = \alpha \circ h$. We need to find unique a $\gamma : U \to Z$ such that $\alpha = \gamma \circ f$, but before we can choose such a γ, we need to do some work. We would like to define $\gamma(u) := \alpha(f^{-1}(u))$ for every $u \in U$ with positive measure, but this is only well defined if α is constant on the fibers of f, which is what we are going to show.

Let $u \in U$ with positive measure. Let f' be a representative of f. Then since f' is measure preserving and u has positive measure, $(f')^{-1}(u)$ is nonempty. We will show that α is constant on the subset of $(f')^{-1}(u)$ consisting of all elements with positive measure. So let $v_1, v_2 \in (f')^{-1}(u)$

such that v_1 and v_2 have positive measure. It suffices to show that $\alpha(v_1) = \alpha(v_2)$. Notice that by Lemma 3.4.7 that there is a finite probability space W with two measure preserving maps $g', h' : W \to V$ such that

$$g' : v_1 \mapsto v_1 \text{ and } g' : v_2 \mapsto v_2,$$
$$h' : v_1 \mapsto v_2 \text{ and } h' : v_2 \mapsto v_1.$$

We will show $f' \circ g' = f' \circ h'$. Since $v_1, v_2 \in (f')^{-1}(u)$, we have that

$$(f' \circ g')(v_1) = f'(v_1) = u = f'(v_2) = (f' \circ h')(v_1),$$
$$(f' \circ g')(v_2) = f'(v_2) = u = f'(v_1) = (f' \circ h')(v_2)$$

and since by definition of g', h' in Lemma 3.4.7, $(f' \circ g')(z) = f(v_1) = (f' \circ h')(v_z)$, where v_1 is assumed to have larger or equal measure than the measure of v_2. We also have by the same definition in the lemma that for all w different from v_1, v_2 and z, we have $(f' \circ g')(w) = f(w) = (f' \circ h')(w)$. So all in all we have $f' \circ g' = f' \circ h'$. Now defining $g = [g']$ and $h = [h']$, we have that $f \circ g = f \circ h$ and therefore, by assumption, $\alpha \circ g = \alpha \circ h$. Then we see that

$$\alpha(v_1) = \alpha(g(v_1)) = \alpha(h(v_1)) = \alpha(v_2)$$

which concludes that α is constant on the fibers $(f')^{-1}(u)$. We can now finally define γ. We do so by defining $\gamma(u) := \alpha(f^{-1}(u))$. This is well defined as α is constant on the fibers $(f')^{-1}(u)$. Now also for every $v \in V$ with positive measure, since f' is measure preserving, there is a $u \in U$ with positive measure such that $v \in (f')^{-1}(u)$ so that $\alpha(v) = \gamma(f(v))$ and therefore $\alpha = \gamma \circ f$.
We now show that γ is unique. Suppose that there is another reduction $\gamma' : U \to Z$ such that $\alpha = \gamma' \circ f$. Then for every $u \in U$ with positive measure, since f is measure preserving, there is a $v \in V$ with positive measure such that $f(v) = u$. Then since α is constant along the fibers of f, we have that

$$\gamma(u) = \alpha(f^{-1}(u)) = \alpha(v) = \gamma'(f(v)) = \gamma'(u).$$

So γ is unique and therefore the atomic topology is subcanonical on **Prob**. □

The proof of Theorem 6.2.1 demonstrates the surprising result that each morphism in **Prob** is actually a so called coequalizer, which is the categorical dual to the equalizer.

Lemma 6.2.2. *Let J be a Grothendieck topology on* ***Prob*** *and let the empty sieve $\emptyset$ be a covering sieve over U. Suppose F is a sheaf. Then $F(U) \simeq \{$**pt**$\}$.*

Proof. Since F is a sheaf, for every matching family $\{x_f\}_{f \in \emptyset}$ of $\emptyset$ for F, there is a unique amalgamation $x \in F(U)$. Now we have $\{x_f\}_{f \in \emptyset} = \emptyset$, so any element of $F(U)$ is an amalgamation. But since x is the unique amalgamation, $F(U)$ must consist of only one element or in other words $F(U) \simeq \{\mathbf{pt}\}$. □

Theorem 6.2.3. *The atomic topology is the canonical topology on* ***Prob****.*

Proof. Let J be subcanonical Grothendieck topology larger than J_{at} on **Prob**. Then there is a J-covering sieve on some object U which is not a J_{at}-covering sieve on U. Then by definition of J_{at}, this J-covering sieve must be the empty sieve. Now we have a morphism $\pi_U : U \otimes \mathbf{2} \to U$ and then

$$\pi_U^*(\emptyset) := \{g \mid \pi_U \circ g \in \emptyset\} = \emptyset$$

and by the stability axiom, the empty sieve is also a J-covering sieve over $U \otimes 2$. Then notice that for every $u \in U$ we have a map $f : U \otimes \mathbf{2} \to U \otimes \mathbf{2}$ defined by

$$f(u, i) = \begin{cases} (u, 2), & \text{if } i = 1, \\ (u, 1), & \text{if } i = 2, \end{cases}$$

and it is seen that $f \neq \mathrm{id}_{U\otimes\mathbf{2}}$. Moreover, this map is measure preserving as $p_{U\otimes\mathbf{2}}(u,1) = \frac{p_U(u)}{2} = p_{U\otimes\mathbf{2}}(u,2)$. We therefore get two different morphisms $[f], \mathrm{id}_{U\otimes\mathbf{2}} \in \mathrm{Hom}_{\mathbf{Prob}}(U\otimes\mathbf{2}, U\otimes\mathbf{2})$ and in particular $\mathrm{Hom}_{\mathbf{Prob}}(U\otimes\mathbf{2}, U\otimes\mathbf{2}) \not\cong \{\mathbf{pt}\}$. Then by Lemma 6.2.2, $\mathrm{Hom}_{\mathbf{Prob}}(-, U\otimes\mathbf{2})$ is not a sheaf and J is therefore not subcanonical. Then J_{at} is the largest subcanonical Grothendieck topology and it is therefore the canonical topology. □

Corollary 6.2.4. *All Grothendieck topologies on **Prob** with no empty covering sieves are subcanonical.*

Proof. This is just an application of Lemma 4.6.11 and Theorem 6.2.1. Since the atomic topology is subcanonical and every Grothendieck topology with no empty covering sieves is a smaller Grothendieck topology than the atomic topology, all Grothendieck topologies on **Prob** with no empty covering sieves are subcanonical. □

So all examples we have given, the X-projection topologies, projection topologies and combinations of them, are subcanonical Grothendieck topologies on **Prob**. Moreover, every Grothendieck topology J on **Prob** is subcanonical if and only if it does not declare the empty sieves to be covering.

7 Discussion

Though Grothendieck topologies have proven their usefulness in areas of mathematics like topos theory and synthetic differential geometry, we may not need them in the theory of finite probability spaces. Since we have not looked at the internal logic of the categories of sheaves that the X-projection topologies and the projection topology induce, we may as well first investigate the finite probability theory inside the presheaf category $\mathrm{Psh}(\mathbf{Prob})$. If all the properites that we care about inside $\mathrm{Psh}(\mathbf{Prob})$ are the same as the properties of **Prob**, there is no need find a Grothendieck topology J so we can restrict down to $\mathrm{Sh}_J(\mathbf{Prob})$.

Moreover, since we already have found the canonical topology on **Prob**, we may not even need to consider other Grothendieck topologies on **Prob**. The canonical topology is the largest Grothendieck topology that allows an embedding $\mathbf{Prob} \hookrightarrow \mathrm{Sh}_J(\mathbf{Prob})$.

There are a lot of possibilities where further research can be conducted. There is a large class of Grothendieck topologies on **Prob** that we have not explored in this paper, so called subcategory topologies. Every full subcategory of a category defines a Grothendieck topology on the category. A possible interesting example would be to look at the full subcategory of **Prob** consisting of all finite probability spaces that have a uniform distribution and which Grothendieck topology it induces. Another interesting example would be to consider the full subcategory of **Prob** consisting of finite probability spaces which have no elements with measure zero.

Another interesting direction would be to conduct the theory of finite probability spaces in $\mathrm{Psh}(\mathbf{Prob})$. In synthetic differential geometry, the theory of manifolds inside some sheaf categories $\mathrm{Sh}_J(\mathbf{Man})$ have a stronger, more generalized version of Stokes theorem. Would theorems in probability theory like the law of large numbers, central limit theorem also hold in the generalized theory of $\mathrm{Psh}(\mathbf{Prob})$? And are there, just like in the case of synthetic differential geometry, stronger theorems that hold in the probability theory for $\mathrm{Psh}(\mathbf{Prob})$?

8 Conclusion

In this thesis, we gave a rigorous foundation for the category of finite probability spaces **Prob** and a characterization for isomorphisms in this category is given which are built upon [1]. We also provided a proof that **Prob** does not admit products.
We then give a brief overview on the theory of sheaves and Grothendieck topologies, following mostly [5]. We start with defining the notions at the heart of sheaf theory such as sieves, Grothendieck topologies, presheaves, matching families amalgamation and sheaves, and prove some elementary properties of them.
Furthermore, we define Grothendieck topologies on **Prob** by using the notion of coverages and give a few simple examples of a coverages on **Prob**. It is then seen what Grothendieck topologies are generated by each of the individual coverages and found the X-projection topologies and the projection topology and it is proved that they are nontrivial. By taking unions we can create larger examples of coverages on **Prob** with the ones we already had and we investigate what the Grothendieck topologies generated by the unions of coverages are.
While proving that the aforementioned Grothendieck topologies were subcanonical, we realised by coincidence that there was a more general result that we could prove using [10]. We realised that the atomic topology was not only a subcanonical topology, but the canonical topology on **Prob**. This very powerful result will prove all that Grothendieck topologies, different from the discrete topology, are subcanonical on **Prob**.

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